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INFRASTRUCTURAL AND DIFFERENTIATED INSTRUCTION IMPLEMENTATION CHALLENGES OF THE SELECTED PUBLIC SCHOOLS

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Over the years, education has changed rapidly and significantly. The physical condition of schools is also a significant factor in the implementation of education policies. Theriana and Al-Amin (2025) stated that infrastructure includes buildings and systems such as classrooms, labs, technology facilities, and internet access that support teaching and learning. In recognition of this, DepEd Order No. 24, s. 2022 adopted the Basic Education Development Plan (BEDP) 2030, which explicitly recognizes the lack of school infrastructure as a major challenge and outlines strategies for addressing these gaps, including the expansion and upgrading of facilities.

The "traditional" one-size-fits-all approach is increasingly considered outdated for addressing the complexities of classroom that includes pupils from diverse backgrounds and bring their own unique skills, challenges, and learning styles (Ab Hajis and Othman 2024). To effectively address this diversity, the differentiated instruction (DI) in teaching approach seeks to tailor the learning individual needs by modifying content, the process, and the assessment (Ferrer and Naanep 2025). Furthermore, in line with Republic Act No. 10533, the Enhanced Basic Education Act of 2013, the Department of Education (DepEd) is required to make the basic education curriculum more learner-centered, inclusive, and developmentally appropriate. Consequently, to meet the needs of all these different students, differentiated instruction is used (Republic Act No. 10533 2013 n.d.)

However, the successful implementation of DI is often hindered by environmental constraints. According Bai (2023); Donkoh and Amoakwah (2024) that lack of resources and infrastructure in schools affects the quality of education students receive, as it can hinder the implementation of differentiated instruction in classrooms. Specifically, Hussain and Afzal (2023) emphasized that the

lack of infrastructure and inadequate educational facilities in public schools, such as overcrowded classrooms, insufficient furniture, and poor sanitation, create an environment inconducive to learning. In addition to these physical barriers, Zou et al. (2025) note that infrastructure issues such as unstable internet connections, a lack of devices, and insufficient technical support remain significant barriers to the effectiveness of digital learning, especially in developing areas.

Donkoh and Amoakwah (2024) stated that public school teachers could not use learner-centered pedagogies because they lacked sufficient teaching and learning resources, their classes were too large, there were insufficient resources across all subject areas, and teachers did not know much about them. Even with these problems, the movement toward inclusive, learner-centered teaching, such as differentiated instruction, continues to grow. This shows how important it is to fix these infrastructure problems right away.

Derder et al., (2023); Hussain and Afzal (2023) highlighted key obstacles that affect the quality of school facilities and resources. These include unequal access to digital tools, inadequate physical infrastructure, and limited technical support systems. Such issues are widely recognized as affecting the overall quality of education delivered in public schools. Despite these documented obstacles, existing literature often fails to explore how these infrastructural challenges specifically affect teachers' ability to implement differentiated instruction (DI) in the classroom. Nevertheless, these limitations are well documented, few have examined their direct impact on instructional strategies. Moreover, little is known about how these issues affect public school teachers in specific local contexts. This study aims to address this gap by investigating how infrastructure-related challenges influence the implementation of differentiated instruction in selected public schools.

Evidence from international contexts further illustrate this struggle. Lawrent (2020); Maldonado and Witte (2020) reported that Tanzanian schools lack basic facilities, such as labs, libraries, toilets, classrooms, playgrounds, water points, and electricity, and combine with unequal resource distribution, leads to poor student results, compromised teacher performance, and elevated teacher absentee rates. Similarly, Daka and Kakupa (2020) stated that implementing differentiated instruction faces particular obstacles in rural schools, as their existing infrastructure is insufficient to meet its requirements.

Adding to this complexity, Saigar and Jamaludin (2025) identify four main problems with implementing differentiated instruction: time and planning, resources and materials, institutional support, and student management. The report in Zambia discusses both the problems and benefits of using differentiated instruction (DI) in high school math classes in Kalomo District. The obstacles teachers faced included limited access to professional development programs, a lack of resources, large student groups, and the rigidity of the educational material, which prevented them from successfully applying DI techniques (Kaziya 2025).

The Philippine context mirrors these global struggles that fundamental infrastructure problems persist, as classrooms are overcrowded and students lack access to proper sanitation facilities, clean water, and reliable electricity services. Consequently, these problems affect students and teachers, making it difficult for them to use differentiated teaching methods, according to Navarro (2022); Kilag et al. (2024). Similarly, Rosario et al. (2024) found that teachers who lack training in differentiated instruction tend to show limited knowledge of differentiation strategies, underuse available school resources, perform ineffective assessment practices, manage classrooms poorly, and mishandle student behavior.

According to Peralta et al. (2025), implementing differentiated instruction was difficult for several reasons. For example, planning lessons took much time, students had behavioral problems, and using different assessment methods was difficult. Lumando et al. (2023) at Tampakan, South Cotabato, showed that differentiated instruction increased students' interest in learning and improved their school performance. The teachers identified multiple obstacles when implementing the new system, as their schools lacked essential resources, required extensive training, and had limited time, and faced resistance from staff members (Ferrer and Naanep 2025).

Public schools face challenges in providing adequate infrastructure and customized instruction due to insufficient funding, resulting in substandard facilities and teachers lacking adequate training in differentiated teaching methods (Olabiyi et al., 2025). It can hinder teachers' ability to implement differentiated instruction effectively, thereby affecting the overall quality of education and equitable access to learning opportunities. This study aims to identify the challenges in implementing infrastructural and differentiated instruction in selected public schools in Nasipit, Agusan del Norte.

The research serves as an important tool in examining the factors that influence both educational quality and equal access to learning in public schools. By focusing on infrastructure and differentiated instruction, the study highlights how physical resources, learning spaces, and instructional strategies directly affect teachers' ability to implement effective and inclusive teaching practices. Understanding these factors provides evidence-based insights into the challenges teachers face in delivering learner-centered instruction, particularly in resource-constrained environments.

The significance of this study extends to multiple stakeholders in the education sector. For teachers, the findings offer practical guidance on how differentiated instruction can be adapted despite infrastructural limitations, thereby improving classroom engagement and learning

outcomes. School leaders and administrators can use the results to make informed decisions on resource allocation, professional development, and school improvement planning. These insights support the creation of learning environments that are conducive to innovation, collaboration, and inclusivity.

In conclusion, the study holds substantial value for policymakers and educational support staff, as it provides empirical data that can guide policy formulation and program development aimed at promoting educational equity. By identifying gaps in infrastructure and instructional support, the research emphasizes the need for targeted investments and continuous teacher training to ensure that all learners, regardless of background or ability, have access to quality education. Ultimately, this study contributes to strengthening the public education system by addressing students' diverse learning needs, fostering inclusive and equitable learning opportunities, and supporting the long-term academic, social, and economic development of communities.

1.2 Literature Review

This chapter discusses the problems that teachers in some public-school face in setting up infrastructure and implementing differentiated instruction. It includes all relevant local, national, and international literature that demonstrates how serious infrastructure problems hamper their ability to provide quality education.

Infrastructural Challenges of public schools

The state of school buildings has a significant effect on how well students learn and how good their education is. There are significant infrastructure problems in the Philippines that make it hard to provide a good education. Poor facilities, such as inadequate classrooms, wash facilities, electricity access, and ICT access, can result in uncomfortable, unhealthy, and distracting physical learning conditions for both children and teachers (Navarro 2024).

Furthermore, this challenge is not unique to the Philippines; for instance, despite their invaluable contributions, Ghanaian mission schools face persistent infrastructural deficits that hinder their ability to deliver quality education. The infrastructure of mission schools in Ghana includes various physical facilities essential for effective teaching and learning, such as classrooms, libraries, laboratories, and administrative offices (Essien-Baidoo 2024). Moreover, the integration technology in schools is challenging due to shortages of equipment, internet access, electricity, and teacher training (Jegede 2019).

Similarly, a study by Ayoko et al., (2023) found that physical facilities for teaching and learning in Nigerian universities are inadequate, substandard, overstretched, overcrowded, and often improvised. An average public university in Nigeria lacks basic infrastructure such as reliable water supply, electricity, and standard accommodation for students.

Inadequate Classroom

Inadequate classroom conditions remain a persistent concern in many educational settings. According to Navarro (2024), although classroom congestion has improved at the national level, spatial inequality in classroom student ratios persist across regions. Many schools, particularly in remote and disadvantaged areas, continue to experience overcrowding and require additional classrooms. This situation is largely attributed to increasing enrollment, deterioration of existing school buildings, and damage caused by natural calamities. These conditions indicate that a significant number of schools still lack safe and adequate learning spaces.

In this context, Inadequate classrooms characterized by a lack of sufficient space, overcrowding that exceeds regulated student-teacher ratios, and unsuitable physical conditions that negatively impact students' focus, comfort, and interaction during class hours. As a result, this

physical deficit critically compromises the learning environment, rendering the educational process ineffective and yielding suboptimal learning outcomes (Kholisah et al. 2024). Moreover, inadequate classroom facilities and resources disrupt classroom management practices, forcing teachers to rush during dismissals due to the need to shift classes, which undermines lesson continuity and classroom order (Impuesto, 2024).

Similarly, Vakili et al. (2024) stated that overcrowded classrooms environment significantly hinder teaching effectiveness and student engagement. Specifically, High student-to-teacher ratios lead to increased stress, heavier workloads, and burnout among educators, making it difficult to provide individualized attention, which can result in frustration. Consequently, this chaotic environment often leads to disruptive behavior, making it challenging for educators to deliver quality instruction. Meanwhile, students often feel disengaged and struggle to participate meaningfully due to distractions and noise. Consistent with these findings Kamara and Dadhai (2020) also reported that many public schools often face challenges such as low ceilings, insufficient furniture, poor ventilation, as well as inadequate furniture layout in the teaching classrooms. These conditions can limit the successful implementation of DI, as they can limit the use of different assessment methods and leave teachers with little time to plan engaging lessons (Peralta et al. 2025).

Futhermore, Fatima et al., (2019) further highlight the challenge faced by teachers face in teaching and learning process, such as the problem of noise, giving tests, getting students' attention, controlling the class, teachers' voices, students' questions, seating arrangements, and getting the class's full attention. It can be hard to teach in classrooms with too many students due to disruptive behavior, limited space, and insufficient resources. As a result, disruptions in teaching

and learning will prevent students and teachers from performing to their full potential (Tshangana et al. 2023).

In contrast, Ahmed et al., (2020) emphasized that an adequate classroom includes furniture, an electric supply, ventilation, and various amenities. Moreover, factors including room size, temperature, lighting, and teaching styles can all contribute to improving students' learning. The availability and teachers' use of physical and advanced, innovative ICT-related instructional facilities and teaching methods draw students' attention to participate actively in classroom activities, ultimately improving students' learning. Similarly, Seherrie (2023) noted that well-maintained infrastructure provides students with safety, comfort, and opportunities to express themselves, thereby fostering a conducive learning environment that supports academic growth throughout their educational journey. Additionally, classroom with flexible seating and easy-to-reach resources that can help students with a range of learning styles and needs. By creating an atmosphere of belonging, respect, and acceptance, well-designed classrooms can help students make friends and build a strong sense of community.

According to Nwokedi (2023) that the classroom environment comprises many parts, including the physical, social, and pedagogical aspects each playing a crucial role in shaping the learning experience. A physically well-organized classroom with appropriate lighting, acoustics, and seating arrangements sets the stage for conducive learning. This environment emphasizes the importance of cultivating a favorable social climate by promoting supportive interactions among teachers, students, and peers. These interactions not only aid learning but also help build students' confidence, alertness, and active participation. It makes the class a more engaging and fun-filled experience, particularly crucial in the digital era, where online learning often lacks these interpersonal dynamics (Joshi et al., 2025).

In relation to physical learning spaces, Afalla and Fabelico (2020) further emphasize that teachers who keep their classrooms clean, large, well-organized, and well-ventilated make it easy for students and teachers to learn and teach. Setting up learning environments that support students can make them feel safe and supported. There are aspects of the social environment that could be improved, especially regarding relationships with peers and feelings of safety. As Cambay and Paglinawan (2024); Gonzales (2024) note, a teacher's ability to plan and lead lessons that prevent students from getting bored, acting out, or becoming inattentive is key to effective teaching. It was also emphasized that a supportive, positive classroom environment, the use of behavior management strategies, and professional development and resources for teachers can help reduce the stress associated with behavior problems.

In addition, Ibrahim and Aslam (2025) stated that classrooms lacking adequate facilities for teaching and learning are difficult to use effectively. Poor facilities, a lack of essential teaching materials, an unstable power supply, and understocked labs can all lead to low student performance, low motivation, and a poor learning environment. In support of this Bresenio (2025) also said that poor or broken lighting and ventilation, unstable armchairs or tablet chairs, a lack of space, cluttered or badly organized display areas, and poorly maintained chalkboards or whiteboards all make it hard to see. Moreover, outdated or lackluster bulletin boards make learning less interesting, and that disorganized cabinets for teaching materials make it hard to find resources. He also said that the demonstration table is not used effectively, which limits hands-on learning and unclear or poorly displayed door labels which leads to confusion.

Electricity Access

Access to electricity remains a critical challenge in many public schools in the Philippines. According to Navarro (2024), although several neighboring countries in East and Southeast Asia

have achieved universal school electrification, the Philippines continues to struggle to complete this goal. Moreover, the study also highlights the need to upgrade existing electrical connections to ensure a stable power supply. Limited and unreliable electricity constrains the effective use of technology and modern teaching tools in schools.

Electricity in education plays a crucial role by providing students with better learning environments and better outcomes. In school, providing electricity has both direct and indirect benefits; the former includes the improvements in lighting and access to any electricity-dependent appliances or equipment, while the latter benefits through enabling internet access, which may allow access to other educational services/applications and content for students and their faculties (Cabotaje 2021). Futhermore, Capule-Navarro et al., (2020) show that having access to electricity can have different effects on education. For example, school electrification extends teaching and studying hours in the morning and/or evening, whereas without it, classes depend on natural light and the season.

In the context of contemporary education, Dharmaraj (2025) emphasized the importance of electricity in modern schools. Particularly, digital learning requires more reliable power, underscoring the importance of students' access to electricity in today's schools. Consequently, providing electricity to all areas is important for educational equality, so that kids in rural areas have the same chances as kids in cities. This electricity enabled the use of information and communication technologies (ICTs), such as laptops and audiovisual technologies, in new ways. These ICTs gave people new ways to do things, such as creating PowerPoint slides and listening to digital recordings (Cabotaje 2021). The progress of technology has made learning easier In addition the advancement of information technology (IT), schools usually rely on standard electrical settings and electronic devices. Additionally, electricity is an important part of modern teaching and learning (Sutoyo and Yudhanto 2023).

Moreover, Langelaan et al. (2024) noted that electricity is necessary for technology-enhanced learning environments that support teachers in effectively implementing Differentiated Instruction (DI), using digital platforms for individualized instruction and ongoing formative assessment. As a result, electronic differentiated instruction makes traditional activities easier and gives each student a personalized learning path that fits their needs. This makes students more engaged and gives them more freedom. Simlarly, Halkiopoulos and Gkintoni (2024) stated that electronic differentiated instruction goes beyond the ideas of diversity by adapting content, processes, and outcomes to each learner's strengths and weaknesses. This makes it easier to find and effectively promote differentiated instruction. Nevertheless, Kótay-Nagy (2022) said that technology can help differentiate learning. However, its effects depend heavily on the resources available, the quality of teacher preparation, and the effectiveness of technology use. In support of this view, Haymon and Wilson (2020) discuss these traditional teaching approaches further, especially in basic literacy skills. They say that combining electronic differentiated instruction with these old-fashioned methods may give students the best overall help.

Conversely, a lack of electricity makes it hard for everyone to access education, especially in rural areas. Mamador et al. (2025) reported that well-trained and educated teachers do not live and work in communities that may need them the most because they lack access to electricity, TVs, computers, and other services. Similarly, Mclaughlin (2025) emphasizes that without access to various teaching aids, electronic learning platforms, and professional training, students will struggle to implement beneficial, student-focused, differentiated education. Therefore, Ensuring access to quality education for communities in rural and/or remote areas is crucial for achieving social equity, particularly for populations that have been historically underserved due to the absence of adequate infrastructure, including electricity access and internet connectivity. (Moner-Girona et al. 2025).

In line with previous studies, Du Plessis and Mestry (2019) reported that many schools lack the necessary infrastructure to be safe, efficient, and effective, and that most rural schools lack electricity in school settings. Consequently, students in rural areas face unstable internet connections, limited electricity, and teachers who are not well-trained to teach in online classrooms. Schools need to ensure that students have access to electricity and a stable internet connection (e.g., Wi-Fi) so that teachers can use digital resources tailored to each student's needs (Padilla Rodríguez et al. 2021, In a similar context, MHLANGA (2020) noted that some groups in South Africa, particularly schools in rural areas, have difficulty connecting to the internet and sometimes lack electricity.

In addition, reliability remains a major problem even when electricity is available. Teachers complained about (a) power outages, as teachers who use Wi-Fi (which requires electricity) were significantly affected because they need power to do their jobs, such as charging their laptops and printing modules (Agayon et al., 2022). Additionally, this subject has not been fully explored, even though it is known that some schools in rural Africa lack access to electricity (Moner-Girona et al. 2025)

ICT (Information and Communication Technology)

In terms of Information and Communication Technology, Navarro (2024) reported that targets for computer package delivery were not met, and in some cases, the proportion of schools with computer packages declined. Compared with neighboring countries, Philippine schools show lower levels of ICT readiness. These limitations hinder the effective integration of technology in teaching and learning.

Within the broader educational context, Integrating Information and Communication Technology (ICT) in education helps sustain teaching and learning even in the face of unexpected interruptions. The enormous flow of information and the use of technology have emerged in all fields worldwide. Accordingly, ICT competency frameworks aim provide teacher educators with a basic set of essential competencies that enable them to integrate ICT into their teaching, assist learners in advancing their learning, and fulfill their professional responsibilities daily. Consequently, the implication is that studying different competency frameworks from diverse viewpoints helps to promote teacher educators' abilities to integrate ICT (Hennessy et al. 2022).

The study of Chen and Liu (2025) examined how students 21st-century skills develop when they use Information and Communication Technology (ICT). The study showed that integrating ICT into the classroom works and makes a significant difference in helping students learn many 21st-century skills, such as citizenship, creativity, and collaboration. Similarly, Sayaf et al. (2022) note that ICT provides learners and teachers with access to vast amounts of information, such as the Internet, which serves as a global repository of knowledge. This facilitates learners' ability to view multiple-perspective ideas, receive current information, and utilize self-directed learning. Moreover, one of the greatest strengths of ICT is its potential to cater to multi-learning needs. Multimedia presentations, interactive simulations, and learning software enhance lessons in terms of their comprehensibility and accessibility (Qaddumi et al., 2021).

In addition, ICT-supported instructional strategies have been shown to enhance learner understanding, inspire students, and increase engagement in the learning process. Specifically, (Bado 2022) reported that teachers were able to try out and study different ways to teach and learn when they used digital games in the classroom. It was also found that teachers who used instructional activities across three stages (before the game, during the game, and after the game)

achieved the most significant learning and student participation. Most of the time, it equips students with the skills they need to succeed in the digital age. Adopting ICT in schools does more than improve lessons; it also gives students the skills they need to succeed in the digital world. Learning about technology and digital tools equips them with skills that will help them in future jobs that increasingly require technological know-how (Gupta et al., 2020). Today, many ICT-related teaching tools are used in classrooms, like computers, smartphones, laptops, tablets, LCD screens, multimedia, and projectors (Shams et al., 2019). Collectively, These tools make teaching and learning more interesting and compelling. Accordingly, the delivery of teaching and the learning process itself. Thus, ICTs in education improve teaching and learning at all levels Ravishankar et al. (2024).

However, the study by Papadakis et al. (2021), the most significant problems in implementing new technologies include teachers' reluctance to use them, insufficient training, and limited resources in the classroom. As a result, many teacher educators lack the knowledge and confidence to use technology effectively in their lessons, an important step toward improving student outcomes (Garba et al. 2020). Consistent with these findings, Ifinedo et al., (2020) noted that Nigerian schools also lack adequate technology infrastructure. There may also be problems or resistance when Information and Communication Technology (ICT) is added to the curriculum. Many colleges lack the necessary technology infrastructure, such as a stable internet connection and up-to-date hardware, making it harder for teachers to use technology effectively. Consequently, the limited availability and underutilization of ICT tools undermine the overall effectiveness of technology integration initiatives (Ifinedo et al. 2020; Love Okon et al., 2025).

Furthermore, this study also shows that students in rural areas do not lack their own devices or reliable Internet access, making it harder for some to participate online. These barriers

make it harder for students without access to technology to connect with those who do (Lembani et al. 2020). Another study reveals that students face challenges with internet accessibility at their location. Both their socio-economic status and their geographical location affected how often they could use the internet and their devices. The availability of the internet in their areas may vary and appear inconsistent, making it difficult for rural high school students to have full access to technology every day (Danao et al., 2024).

Differentiated Instruction

Ab Hajis and Othman (2024) discuss the three main components of differentiated instruction: content, process, and product. They also discuss their roles and how important they are in meeting the needs of all students. Specifically, the study focuses on the theoretical foundations and challenges of effectively implementing these variables. Moreover, It emphasizes that well-adapted classrooms create a welcoming environment where all students can reach their full potential. Although the study's specific empirical findings on the effects of these sub-variables are more theoretical. They suggest that targeted strategies for differentiating content, process, and product are necessary to address diversity issues in the classroom and to make teaching more effective.

Similarly, the implementation of differentiated instruction is very essential in order to address diverse needs of students. For instance, Mardhatillah and Suharyadi (2023) include students' readiness, interests, and learning profiles. Thus, teachers can effectively use DI throughout instruction, improving teaching, support, and learning for all students. By recognizing the distinctions among students, it is crucial to enhance both teaching and student learning. However, teachers face challenges including limited awareness, insufficient training, difficulties in classroom management, and resistance to change (Saikia and Ranjan 2025).

Content

According to Ab Hajis and Othman (2024), differentiating content involves adjusting what students learn based on their readiness levels, interests, and learning profiles. Teachers may vary instructional materials, texts, or examples to ensure that all learners can access key concepts. However, the selecting and preparing appropriate content for diverse learners can be challenging, particularly when teachers lack sufficient resources or training. These difficulties often limit the consistent application of content differentiation in classrooms.

Content includes topics, ideas, knowledge, and the curriculum students use to learn. Different students learn and process information in different ways (Del Mar 2024). Roiha and Polso (2021) stated that content can be changed to fit those needs. Content also includes information, skills, and attitudes related to the subject and the materials, as well as the way learning occurs. Similarly, Celik (2019) Teachers are the ones who create, design, and develop their own teaching styles and methods to contextualize the subject to the actual teaching and learning conditions of their classes. Specific styles and tools are employed to present the content of each subject, as outlined in the curriculum. Teachers employ similar methods and styles of teaching, even though topics and content differ within a specific subject, and fail to consider individual differences among learners (Peralta et al. 2025).

According to Irdhina et al. (2021) that in differentiated learning, there are three ways to create different lesson content, namely: 1) Adjusting what the teacher will teach or what students will learn based on the level of readiness; 2) Adjusting what will be taught by the teacher or what will be learned by students based on the interests of students; and 3) Adjusting how the content to be taught or learned will be conveyed by the teacher or obtained by students based on the

learning profile owned by each student. It is assumed that in this study, the mover teachers' content follows each student's learning profile.

The study of Kharka and Kinley (2024) shows that content is changing textbook content, using a variety of teaching tools, creating respectful tasks, checking whether students are ready for lessons, and adapting lessons to help students who are learning slowly. Teachers adapt the content using different teaching tools, such as PowerPoint presentations, video clips, and printed handouts. Suryati et al., (2024) identify two strategies: 1) using varied text and resource materials, where the teacher's previous reliance was solely on textual materials for teaching. Under the differentiated approach, the teacher adapted the learning materials and sources to cater to the varied needs and interests of the students. 2) varied support systems as the second strategy for differentiating content. To accommodate students' diverse learning styles, such as visual, auditory, and kinesthetic, the teacher incorporated a range of audio and video resources alongside traditional print materials.

Similarly, Hasnavati (2022) stated that content involved teachers selecting different learning materials from various sources and combining them in different ways, such as video presentations, PowerPoint slides, or PDF learning modules. Kristanti (2024) emphasized the visual, auditory, and kinesthetic approaches. Where each student has a different level of understanding and learning style, which can increase their ability to complete learning and improve learning outcomes, some other possibilities for differentiating content are: using content at varying difficulty and complexity levels; using assignments' lists at readiness levels of students; presenting ideas through both auditory and visual means; meeting with small groups to re-teach an idea or skill for struggling learners, or to extend the thinking or skills of advanced learners. Regardless of the strategy employed, all students must be aware of their responsibility to meet the specified deadlines for class projects.

Differentiated instruction in content delivery means that different teaching methods are used for each subject that the respondents are responsible for. Differentiated instruction means that teachers use a variety of teaching materials, allow students to learn at their own pace, and form groups to encourage active participation. These actions show that teachers create many learner-centered learning opportunities and adapt to each student's needs, which encourages active participation in the teaching and learning process. The instruction shows that the connections made by the respondents are not very useful and need to be elaborated on in depth so that students can have meaningful learning experiences (Peralta et al. 2025).

Furthermore, based on the findings of Onyishi and Sefotho (2020) the failure to differentiate instruction by teachers is attributed to several factors, including time constraints, a lack of diverse learning facilities, rigidity in the school population, and inadequate knowledge and skills among teachers. Tajik et al., (2024) stated that Afghan teachers face a lack of resources, which hampers their ability to differentiate content. Higher education administrators should address this issue so that teachers can implement content differentiation in their classes tailored to their students' needs. Numerous teachers noted that, due to the constrained instructional time, they were obliged to select and adapt educational content and materials accordingly. In particular, they could simplify or reduce the learning content to ensure students' learning (Yuan and Zhang 2022).

Sulistianingrum et al. (2023) note that content is affected by many factors, including the teacher's education, classroom teaching experience, external factors such as facilities and infrastructure, and time. Hatmanto and Rahmawati (2023) found it challenging to differentiate content for teaching reading skills. Their research shows that many significant problems make it hard to do things like give students access to a variety of resources, make sure that content is

aligned with curriculum standards, deal with time constraints, figure out what each student needs, adapt text complexity, and give each student individualized instruction while keeping content interesting and up-to-date.

Moreover, Celik (2019) emphasized that teachers should adjust their expectations and adapt the learning content and activities to their students' abilities and differences. This is not a disservice to the students when they are given meaningful and appropriately differentiated content that will gradually develop their knowledge. Students appreciate, feel more relaxed, and participate more willingly when the learning content aligns with their level of reasoning. Teachers can integrate content that involves the integration of subject areas from diverse traditions, and it may even emerge from students' own questions, allowing them to construct their own meaning.

Karim et al. (2025) stated that educators must strategize and arrange materials, tasks, and activities to be carried out either in the classroom or remotely, as well as culminating assessments, all customized to accommodate the varying readiness levels, interests, and preferences of their students. This approach encourages teachers to provide various learning opportunities and support systems, such as offering different levels of difficulty in assignments or providing alternative learning materials, to cater to students' individual strengths and areas for growth. By embracing differentiated learning, educators empower students to take ownership of their learning journey while ensuring that all learners have equitable access to educational opportunities and resources.

Process

Ab Hajis and Othman (2024) explain that process differentiation focuses on how students engage with and make sense of learning activities. This approach requires teachers to design varied

instructional strategies, such as flexible grouping, tiered activities, and guided practice. The authors emphasize that managing multiple learning processes simultaneously can be demanding, especially in large and diverse classrooms. As a result, teachers may struggle to effectively implement differentiated processes on a regular basis.

Content encompasses the process of how students learn involves adapting teaching methods, group activities, strategies, and learning activities to meet the students' needs. All students need to achieve the same learning goals, but teachers can differentiate the way students learn the content. There are various ways to acquire knowledge and gain skills, including small groups, instruction, partner projects, and other supported learning methods (Del Mar 2024). DI in secondary schools enhances student learning by tailoring lessons to individual readiness, interests, and learning styles. However, it requires teachers to possess adequate training, preparation time, and collaboration with peers to apply differentiated methods effectively (Smale-Jacobse et al. 2019).

According to Jager et al. (2022) the operationalization of adaptivity and proactivity requires several perspectives: the teachers' thought processes during planning of a lesson are needed to identify the proactivity, and information on children's experiences and (changes in) academic achievements is needed to identify whether teachers' adjustments indeed match children's needs. In support of this, Pozas et al., (2021) reported that the DI process, which included planning, active monitoring, and reflective change, had a positive effect on students' health, social inclusion, and academic self-concept.

In addition, Suryati et al. (2024) stated that in terms of learning processes, the teacher employed a variety of instructional models to support the diverse learning styles present within the classroom. The teacher facilitated peer learning by organizing students into groups, allowing

students to share knowledge and insights. This collaborative approach was further enhanced by employing strategies centred around multiple intelligences and flexible grouping, ensuring that instructional processes were inclusive and adaptive to the varied needs of the students. In line with this Kharka and Kinley (2024) noted that differentiating processes by teachers offered flexible grouping options and incorporated multiple modes of delivery into one lesson, such as audio, visual, and audio-visual materials. The claim that teachers make regarding membership choices for flexible grouping conflicts with observations and student feedback. In contrast, observation and student feedback revealed that teachers do not offer a choice in membership for flexible groupings, and activities are restricted to predefined groups.

Similarly, teachers can provide appropriate instructions and learning opportunities for each student during the learning process. In particular, continuous assessment during this process will help the teacher understand whether each student has learned to the best of their abilities. To achieve this, teachers must understand the interests, abilities, and level of knowledge of each student. For example, some students can learn well if they listen to verbal or audio instructions, whereas other students, simply listening orally is not enough; they also need to read the explanation in the book repeatedly. Moreover, some students can learn effectively through specific learning-related objects, while others are accustomed to studying alone, learning online, or in collaborative and group settings. Therefore, teachers must be able to understand the needs of each student at the beginning of learning in order to create a different learning process and help each student to learn effectively and have fun. Learning that is appropriate for teachers to apply is the ability to convey ways to solve problems, then apply the process and continue to provide support as students' learning develops (Kristanti 2024).

The learning process considers how students learn skills, strategies, and concepts. Differentiated instruction processes can be tailored in two ways: task differentiation and tool differentiation. Specifically, task differentiation means that students are invited to complete different activities, projects, or tasks. In contrast, tool differentiation provides students with varied resources and materials that cater to diverse learning styles and needs (Puzio et al., 2020). Moreover, the process involves interaction between students, educators, and learning resources in a learning environment where teachers and students exchange information. It helps students learn effectively, acquire knowledge and mastery of skills, as well as develop character and form attitudes and beliefs Qorib (2024).

For Instance, Suryati et al., (2023) stated that the challenges faced by teachers include the time and effort required for preparing and designing the learning process, as well as a lack of support from fellow teachers. However, Noor and Golzar (2024) stated that while instructors recognized the importance of tailoring instruction to meet the diverse needs of the students, many struggled with effectively applying DI strategies in practice. Key challenges included insufficient training and understanding of principles, as well as large class sizes, and limited time for lesson preparation, which hindered teachers from designing practical activities and assessments. Gibbs (2023) identifies key process barriers to implementing differentiation in Australian secondary schools, notably limited planning time, insufficient professional development, and uneven leadership support that prevent DI concepts from becoming sustained classroom practices.

According to Tajik et al. (2024) teachers' knowledge, lesson planning, and classroom management all significantly impact how well DI works, especially when teachers understand their students' learning differences and use flexible teaching methods. Furthermore, Krishan and Al-rsa'i (2023) found that teachers play a crucial role in adapting their teaching strategies to integrate

technology and effectively address learners' diverse needs. Teachers successfully modified their instructions, and students demonstrated higher levels of engagement, participation, and intrinsic motivation to learn, highlighting the effectiveness of teacher adaptation. In support of this, Bagus (2025) revealed that both teachers' instructional competence and adaptability skills significantly affect students' classroom engagement. This suggests that when teachers demonstrate flexibility, responsiveness, and adaptive strategies in their teaching process, students become more attentive, motivated, and participative in class.

Product

In terms of product differentiation, Ab Hajis and Othman (2024) describe it as allowing students to demonstrate their learning through varied outputs or performance tasks. These products may differ in format, complexity, or mode of expression, depending on learners' strengths and abilities. The authors point out that evaluating diverse student products requires flexible assessment strategies and clear criteria. Without sufficient time and assessment support, teachers may find product differentiation difficult to sustain.

A product is how students demonstrate what they have learned after a lesson. Teachers may ask students to demonstrate mastery of a skill or concept through various assessments, such as multiple-choice tests, presentations, or essays, or other methods (Del Mar 2024). Students can demonstrate what they have learned through different kinds of work, such as projects, presentations, or written assignments (Saikia and Ranjan 2025). The products they create demonstrate their learning in various ways, such as through projects, written tests, presentations, and performances, tailored to their abilities (Ginja and Chen 2020).

Teachers' creativity in creating alternative products or evaluation methods (Argyropoulou et al., 2021). As a result of the study, students with other interest tendencies, such as creating interactive PowerPoint presentations, video editing, and narration, among others, cannot be accommodated. The existence of assignment instructions in the form of a single type of product, including the causes of students' creativity, cannot be developed optimally. Therefore, teachers need to understand the diversity of student characteristics so that the planning and learning process are in accordance with the needs and characteristics of students Solihah and Mukaromah (2023).

Based on the findings of Kharka and Kinley (2024) teachers differentiate products using a variety of assessment methods, including presentations, quizzes, project assignments, tests, debates, and project-based and practical work. However, during class hours, the oral mode of assessment is more common. A constrained number of possibilities, including visual, kinesthetic, musical, textual, spatial, creative, and practical, were provided for students to demonstrate their progress. It was also clear that teachers expect their learners to primarily exhibit their work in writing or through practical applications. Similarly, Saikia (2025) that teachers may ask students to demonstrate mastery of a skill or concept through various assessments, such as multiple-choice tests, presentations, essays, and other methods. Moreover, products they create demonstrate their learning in various ways, such as through projects, written tests, presentations, and performances, tailored to their abilities (Ginja and Chen 2021).

Helmi et al., (2023) stated that the teacher's perception was crucial in implementing differentiated instruction in the product. For this reason, teachers need to have the correct perception of differentiated learning. The correct perception will cause the implementation of differentiated learning to run effectively. When focusing on the products of learning, the teacher

empowered students by offering them the freedom to choose how they would complete and present their tasks, whether through individual assignments done outside of class or in-class activities. The teacher employed varied questioning strategies to cater to the students' specific needs and learning profiles, resulting in the implementation of three main activities: student worksheets, group tasks, and presentations. These activities were designed to assess students' understanding and application of the material and to accommodate and celebrate their unique ways of learning and expressing knowledge (Suryati et al. 2023).

Product is a method used by teachers to determine the level of understanding of the material or teaching materials prepared for each student to produce a work (product) from the learning results. To determine mastery of a material, a teacher can administer a test, and students are then asked to write a report on a topic related to the lesson material. The best method is one that suits each student's ability level, interest, and preferred learning style. For example, testing kinesthetic learners involves a practical assessment, while testing auditory learners involves a verbal assessment (Kristanti 2024).

Additionally, Hadi et al., (2020) stated that there are three strategies that teachers can use. Firstly, content differentiation involves presenting material in accordance with students' readiness, interests, and learning profiles. The teacher empowered students by giving them the freedom to choose how to complete and present their tasks, whether individually outside of class or through in-class activities. By using varied questioning strategies tailored to students' needs and learning profiles, the teacher implemented three main activities: student worksheets, group tasks, and presentations. These activities assessed students' understanding and application of the material while accommodating and celebrating their unique ways of learning and expressing knowledge (Suryati et al. 2024).

The study of Massey (2025) identified several key challenges and adverse outcomes related to implementing product differentiation, including limited professional development and insufficient resources, which further hindered teachers' ability to implement differentiated products effectively. Large class sizes made it challenging to provide personalized feedback and monitor multiple forms of student output, resulting in inconsistencies and a reduction in instructional quality. Additionally, these challenges often resulted in a narrower range of product options, limiting student engagement and the potential benefits of differentiated instruction.

Moreover, the analysis revealed that teachers had difficulties integrating specific components of differentiated products. As contended by the teachers, in addition to lacking knowledge and skills of DI, they need to be more aware of how to blend and integrate the two respective curricular components into their daily plans when both syllabi are implemented simultaneously. However, the observation and lesson plan documents revealed that, unlike the content materials and process activities, differentiation of products was not noticed to be significant from the observations. There was less flexibility in product differentiation because the ways students could demonstrate their learning were very limited in almost all lessons. (Shareefa 2021).

Traditional assessment methods (e.g., exams and quizzes) may not be suitable for evaluating the skills developed in a PBL environment, such as teamwork, problem-solving, and creativity. Developing new, practical assessment tools that measure both the process and the product of projects is essential but challenging (Shanmugapriya 2025). Teachers who employ differentiation strategies, such as flexible grouping, tiered assignments, and culturally responsive teaching, are better equipped to support the diverse needs of their students. These strategies

enable teachers to modify content, process, and product to ensure that all students can access and engage with the curriculum effectively (Sabir et al., 2024).

Teachers can provide different ways for students to demonstrate what they have learned or understood from the lesson. This may include strategies such as choice boards, which allow students to choose how they demonstrate their knowledge, project-based learning that provides a real-world application of their knowledge, or peer tutoring to support the overall learning process. Differentiating the product helps students engage in creativity and critical thinking while demonstrating an understanding of the subject matter (Voyager 2023). Differentiation strategies, such as tiered assignments, flexible classification, and varied instructional materials, effectively engage all students. These strategies recognize the diverse learning profiles of students and ensure that instructional content, process, and product are adapted to meet their specific requirements Hatmanto and Rahmawati (2023).

Implementation Challenges of Differentiated Instruction

Shareefa et al. (2019) categorized teachers' definitions of differentiated instruction into three main aspects: using different strategies, addressing student diversity, and promoting student learning. However, the study also identified six significant implementation challenges: time constraints, limited resources, lack of knowledge, large class sizes, inadequate support, and heavy workload.

Most schools do not have access to extensive programs, due to limited resources or other considerations (Sims & Fletcher-Wood, 2021). According to Kahmann et al., (2022) conducted a meta-analysis providing an overview of characteristics and effectiveness of studies focused on DI and revealed a medium effect of such programs on teacher measures. They furthermore exposed

a need for more explicit statements about the programs theory of improvement to support effective evaluation and enable better-informed program adjustments. Insorio (2024) found that differentiated instruction does improve student engagement and success in math. However, it is hard to use consistently due to limited resources, limited time, and too many students in the classroom.

Additionally, Saigar and Jamaludin (2025) found that teachers in rural schools struggled with limited planning time, lack of instructional materials, minimal institutional support, and difficulties managing diverse learners, all of which hinder effective DI implementation. Similarly, Maphumulo and Biccard (2024) revealed that teachers in mainstream primary schools faced challenges related to insufficient time, inadequate resources, overcrowded classrooms, and lack of parental support, leading to increased workload and reduced instructional quality.

Time constraints

According to Shareefa et al. (2019), teachers often struggle with limited time to implement differentiated instruction effectively. Planning lessons that address the diverse needs of students requires additional preparation, which can conflict with regular teaching schedules. The authors note that this constraint may lead educators to simplify instruction or use uniform strategies, reducing the potential benefits of differentiation. Consequently, time pressure can negatively affect the quality and consistency of differentiated learning experiences.

Broadly defined, time constraint refers to the inadequacy of time to complete tasks, such as teaching and pursuing syllabi, that require classroom time, in addition to attending various time-consuming courses. In addition, teachers feel the burden of their duty, as well as the time spent on continuous work at school (Azira Jose et al., 2021). This is compounded by the fact that teachers

have always taken on multiple roles, which can lead to hectic workdays and limited time to complete tasks (Marshall et al. 2022). In response to these pressures, recognizing the needs of the time and setting educational goals and objectives accordingly is the core responsibility of governing bodies to acquire the desired outcomes (Akram and Yang 2021).

Similarly, Hasbullah et al. (2023) stated that effective time management can lead to a healthy lifestyle and a positive mindset without the burden of stress. Managing time effectively involves utilizing tools and methods or techniques to plan and schedule time appropriately. However, a significant issue remains where the time required for lesson planning, assessment, and other teaching-related activities (which are generally carried out beyond school hours) is not adequately captured to assess teacher workload (Yezer 2020). Additionally, new studies examine the intersection of time management, well-being, and professional identity among teachers in developing contexts (Villanueva and Dela Cruz 2022). One practical example of addressing this is found in the Merdeka Curriculum, which uses compound learning. This curriculum focuses on essential content, allowing students to explore concepts and strengthen competencies with sufficient time. In addition, the Independent Curriculum is a crucial part of efforts to restore learning from the crisis that we have long experienced (Nurani et al., 2022).

Despite such curricular efforts, teachers who use differentiated instruction often encounter challenges such as insufficient knowledge and resources, the need for professional development (PD), and time constraints when incorporating these methods (Whitley et al. 2019). Specifically, time constraints act as a challenge to DI, highlighting how limited instructional and planning time can impede teachers' ability to differentiate effectively. He also reported that studies often emphasized the difficulties teachers face in addressing diverse learning needs, such as a lack

of time for planning, unmotivated students, class size, and concerns about behavior management (Gibbs 2023).

Reinforcing these findings, the study of Ziernwald et al. (2022) highlight the challenges of educators in implementing DI, stemming from insufficient time, resources, knowledge, and training. Porta et al. (2024) also found that limited time and large class sizes made it hard to use differentiated instruction (DI) effectively in South Australia. Similarly, Unal and Bodur (2022) found that teachers often struggle with the demands of DI, especially when they must differentiate while managing the limited time available for teaching and planning. Furthermore, when DI is used alongside a required curriculum, more problems arise, such as excessive information, strict curriculum requirements, limited time, prioritizing standardized tests, rigid curriculum, and difficulties in adapting it (Grecu 2023).

According to Nurlankyzy and Elmira (2023), teachers reported that planning for differentiated instruction took much time because they had to adapt lessons to meet each student's needs. Along the same lines, Kamarulzaman et al. (2024) said that planning and implementing differentiated lessons can take a long time, and teachers may not have the time to do them well. For teachers who also have to grade, plan lessons, and work on their own professional development, this can be especially hard. In the context of language education, planning and setting up differentiated instruction (DI) lessons takes a lot of time and effort, which is another problem teachers face when they try to use it in their English classes. Ultimately, DI is quite challenging to implement because it requires preparing learning according to the interests, talents, and abilities of each student. As a result, they needed a considerable amount of time to think and create teaching innovations in preparation for their English lesson with DI (Rosadi et al., 2024).

Moving beyond simple scheduling, Creagh et al. (2023) suggest that "time poverty," as the product of the interplay between workload and work intensification, encapsulates the experience of work along two different axes or vectors. These functions work together, but in different ways, to provide the subjective accounting of time and work that each experience. This review highlights a dissonance between the kind of teacher a teacher wants to be and the type of teacher they have time to be. For instance, Shah et al. (2020) found that some teachers were unable to find the time and the feasibility to apply ICT in teaching due to limited time and a tight working schedule. ICT is seen as helpful in making teaching more efficient and creative, but time constraints hinder its full utilization. Thus, most individuals often face challenges in testing their limits, especially when faced with time constraints and verbal pressure.

Finally, the pre-survey results of the study by Estaiteyeh and DeCoito (2023) identified several themes in the responses of Teacher Candidates (TCs) regarding problems they believed might hinder the implementation of DI. Eight of the 17 TCs talked about the time they needed to prepare, seven about problems with resources, seven about administrative issues like support, funding, class size, and professional development, five about student issues like interest, engagement, or special needs, four about teacher knowledge or skills, three about online teaching during the pandemic, and one about curriculum requirements. Essentially, the results came from the teacher and the school after they reviewed the problems that arose when they tried to implement differentiated instruction. This confirms that teachers struggle because it takes time and effort to plan and set up the learning process (Suryati et al. 2023). In conclusion, Griful-Freixenet et al. (2021) examines the most significant problems teachers cite that prevent DI from being used, reiterating that these problems include not having the right resources, not having enough time to plan lessons, and not being able to plan for DI.

Limited resources

Shareefa et al. (2019) emphasizes that limited access to teaching materials, technology, and instructional tools can impede the implementation of differentiated instruction. Teachers may find it challenging to design alternative activities, provide varied learning materials, or adapt assessments to meet the needs of all students. Without adequate resources, differentiated instruction becomes difficult to sustain in classrooms, particularly in under-resourced schools. This limitation often forces teachers to revert to conventional teaching methods.

In a general sense, limited resources refer to the lack of essential materials, funding, facilities, or workforce needed to support effective teaching and learning. However, limited resources continue to pose a significant challenge to quality education. For instance, Gibbs (2023) found that teachers in Australian public schools often reported that limited photocopy budgets, a lack of teaching materials, and insufficient time were the most significant factors preventing them from differentiating instruction. Similarly, McMahon (2019) emphasized that teachers who wanted to use DI strategies often face problems due to insufficient time or resources. This shows how hard it is to find good and ready-made differentiated resources. Ultimately, these new ideas show that teachers' inability to plan, prepare, and teach differentiated lessons well is directly affected by the resources they have access to.

Conversely, research suggests that Doan et al. (2025) the availability of teaching resources and technology has a significant influence on instructors' capacity to teach diverse courses effectively. Similarly, Grecu (2023) found that differentiated instruction (DI) is more effective when teachers access digital tools, adaptable learning materials, and ample classroom supplies that make it easier for students to participate in a variety of learning activities. This observation holds true in specific regional contexts as well; in the Philippines, Prudente et al. (2024) found that

teachers are much better able to adapt their lessons to meet the needs of all of their students when they have access to professional development programs and enough resources. On the other hand, Insorio (2024) found that teachers do not use new instructional strategies such as DI because they lack the time, money, or teaching materials. Additionally, Doan and Shapiro (2023) note that schools with consistent instructional support and professional learning resources were better able to implement DI than schools that lacked sufficient resources.

The struggle is particularly acute in marginalized areas, as Carrete-Marín et al., (2024) reported that educators in rural and multigrade schools encounter difficulties in producing and obtaining suitable teaching materials due to tight budgets and inadequate infrastructure, compelling them to rely on improvisation that often compromises instructional quality. Similarly, Maphumulo and Biccard (2024) highlighted that teachers in rural and peri-urban regions face numerous challenges, including inadequate classroom resources, large class sizes, and time constraints, which hinder the comprehensive implementation of DI. These trends are further supported by research showing that overcrowded classrooms diminish opportunities for small-group instruction, thereby restricting teachers' abilities to adapt learning tasks according to individual student readiness (Vakili et al. 2024).

Beyond physical materials, systemic issues also play a role; Steiner (2024) noted that numerous school districts lack consistent access to high-quality, curriculum-aligned instructional materials, forcing teachers to create makeshift content. This fragmented system intensifies inequity and reduces the effectiveness of DI. Furthermore, Sarah Martin (2024) reported that inadequate funding for special-needs education in Australia compels schools to reallocate budgets, straining available resources and reducing their capacity to implement differentiated

approaches for diverse learners. These narratives highlight the systemic nature of resource limitations, which have far-reaching repercussions for classroom practices.

Looking specifically at the Philippine context, Hatmanto and Rahmawati (2023) state that while Filipino educators generally appreciate DI, they face several hurdles, including insufficient teaching materials, a lack of professional development opportunities, and inadequate administrative support. Corroborating this, Insorio (2024) found that although differentiated instruction has a positive impact on student engagement and success in mathematics, practical challenges such as insufficient resources, lack of time, and overcrowded classrooms hinder its consistent implementation. This is further validated by the policy analysis released by the Philippine Institute for Development Studies, which suggests that uneven space and poor infrastructure make it hard to provide fair teaching conditions, which, in turn, make it harder for teachers to differentiate (Navarro 2024).

Moreover, Wilson and Schulke (2025) highlighted that deteriorating or unsafe physical environments such as school grounds and facilities contribute to teacher burnout and student disengagement, complicating differentiated teaching efforts. Ab Hajis and Othman (2024) further emphasized that the availability of resources, administrative support, and professional training are interconnected factors essential for creating a supportive environment for DI success. Together, these studies reveal a consistent pattern across various contexts and time periods: limited financial, material, and infrastructural resources consistently hinder teachers' abilities to implement differentiated instruction effectively. In light of these findings, to enhance inclusive teaching practices within public education systems, it is crucial to address these shortcomings through strategic investments in school infrastructure, the equitable distribution of materials, and ongoing support for teachers.

Lack of knowledge

According to Shareefa et al. (2019), a lack of knowledge or insufficient training in differentiated instruction poses a significant barrier for teachers. Many educators are unfamiliar with the strategies required to adjust content, process, and assessment for diverse learners. The study highlights that professional development and ongoing support are crucial for building teacher competence in differentiated instruction. Without such training, teachers may feel unprepared to meet the varied learning needs of their students.

In its most basic form, lack of knowledge refers to the absence or deficiency of understanding, information, or awareness about a particular subject or topic (Pandey 2021). Similarly, Halse (2022) said that knowledge development requires institutional transformation to foster multicultural and diverse learning environments. Beyond institutional shifts, incorporating prior knowledge into instruction enhances conceptual understanding and higher-order thinking skills, further underscoring its role in effective learning (Shalikhah et al. 2023). Furthermore, Hasanah et al. (2023) suggests that teachers' concern and knowledge about the factors that influence student learning outcomes can improve the quality of educational services they provide to their students.

Building on this foundation, Hasanuddin et al. (2024) highlights the significant role of prior knowledge in differentiated instruction, emphasizing that it serves as a foundation for personalized learning experiences. Prior knowledge helps educators tailor instruction to students' individual needs, enhancing engagement and comprehension. When effectively utilized, it fosters meaningful connections between new and existing knowledge, allowing students to progress at their own pace while developing self-regulated learning (SRL) skills. Parallel to this, Gibbs (2023) found that school leaders who have a deep understanding of DI facilitate whole-school

implementation by supporting data-driven teaching, respectful student-teacher relationships, and collaborative teacher learning communities. This implies that developing teacher competence for DI requires considering teachers' knowledge, skills, and attitudes. In short, implementing DI requires teachers to know when and how to teach specific content effectively and to have knowledge of approaches that enable pupils to be taught effectively (Langelaan et al., 2024).

Practical application remains a hurdle, however, as when teachers integrate knowledge about DI into their existing knowledge frameworks about teaching and learning, this can strengthen DI implementation. Since the transfer of knowledge about DI to practice is often challenging, it is recommended that teacher development programs pay explicit attention to the implementation process (Van Geel et al., 2022). Specifically, teachers need to understand their students' strengths and weaknesses, as well as the best methods to support different learners. To achieve this, continuous learning and professional development are essential (Saikia & Ranjan, 2025). Conversely, Nketsia et al. (2024) emphasized that the lack of exposure to the comprehensive model of DI in teacher education programs results in pre-service teachers having inadequate knowledge and an in-depth understanding of its conceptual principles and specific practical strategies.

Furthermore, Mengistie (2020) revealed that teachers were not regularly differentiating instruction in their classrooms due to a lack of knowledge of specific strategies, time constraints for preparing differentiated instruction, and a lack of available resources. Adding to this, Yavuz (2020) shows that another issue in implementing differentiated instruction is teachers' lack of knowledge; they do not have sufficient knowledge about the importance of differentiated instruction, and some of them cannot even describe the term "differentiated instruction". The undeniable fact is that if teachers lack sufficient knowledge and exposure to using these strategies,

they are less likely to use differentiation in their teaching, even though they have been in the profession for an extended period (Shareefa, 2021).

Interestingly, this lack of knowledge has been identified as an issue faced by teachers even when implementing student-based assessments in their respective schools. This study is about observing if this problem (together with selected Moderating Variables) affects teachers' comfort levels to such an extent that they decide to adopt the silent behavior rather than communicate with the management. In many cases, lack of knowledge is due to lack of training and exposure to the new policy (Sethupathy, 2020). Similarly, teachers, particularly those in Islamic Religious Education, face challenges in implementing differentiated learning areas. The challenges include a lack of knowledge about differentiated learning strategies, difficulties in identifying student needs, and a lack of time and resources (Tiara Nove Ria and Kurniati 2023). Furthermore, educators revealed that DI is not consistently applied in inclusive classrooms. While teachers recognized the importance of DI, and occasionally used strategies such as graphic organizers, group activities, and tiered lessons, real challenges arise during the implementation process. Their efforts are hindered by insufficient training, inadequate preparation time, and limited professional development opportunities (Benjamin and Panesar-Aguilar 2020).

Nevertheless, some researchers remain optimistic. Magableh and Abdullah (2020) note that although there are a few challenges in differentiated instruction implementation like planning, the class size, the number of students in the classroom, the shortage of support from administrations and the lack of professional development, the interviewees were positive that these challenges can be minimized and reduced with professional development and more knowledge about differentiated instruction strategies. In support of this, Alsamiri et al. (2023) found out that primary school teachers demonstrate a general understanding of differentiated

instruction, specifically in terms of support for students with varying needs (i.e., fast learners, medium learners, and slow learners). However, teachers often lack knowledge of specific strategies to manage mixed-ability classrooms effectively. Ultimately, Unal et al. (2022) reported that teachers encounter multiple barriers to implementing DI, including insufficient DI knowledge, inadequate practice opportunities, and a lack of resources and professional development.

Large class sizes

Shareefa et al. (2019) note that large class sizes make it difficult for teachers to provide individualized attention and implement differentiated strategies effectively. Managing multiple students with diverse learning profiles simultaneously can be overwhelming. The authors point out that overcrowded classrooms limit opportunities for flexible grouping, personalized feedback, and varied instructional activities. This challenge often leads teachers to adopt one-size-fits-all teaching approaches.

In this regard, class size is often considered one of the crucial factors that determines the effectiveness of teaching and learning in the classroom setting. Large classes are challenging not only for the teachers but also for the students. Specifically, students often lack exposure or struggle to concentrate on the content due to classroom noise, and teachers are unable to provide as much guidance as they would like to. Consequently, teaching and learning cannot become effective in large classes (Chand 2023). This challenge is pronounced in higher institutions, especially for core courses that every student in the department or faculty must take. For instance, over 1,000 students often converge in a small lecture hall with a capacity of 300, making it extremely difficult for lecturers to communicate effectively with students, and this has contributed to the declining standard of education in Nigeria (Gbenga et al., 2023).

In contrast, some researchers argue that these difficulties can be mitigated through pedagogical adjustments. Sholah (2020) emphasize that successful DI in large classes necessitates establishing clear routines, utilizing various pacing techniques, and empowering students with individualized tasks to streamline management. Similarly, Yahya et al. (2024) found that larger classrooms can actually facilitate interactive learning when teachers accommodate different classroom dynamics with individualized attention and tailored materials. In addition, Senyagwa (2021) argues that adopting effective teaching methods is imperative, positing that skilled teachers, a student-centered approach, and adequate materials are the pillars for resolving significant class-size challenges within a DI framework.

However, the psychological and behavioral toll remains a significant concern. Smith and Roberts (2023) examine how managing a large number of students simultaneously can lead to increased behavioral issues and disruptions. Teachers often find it challenging to maintain order and ensure that all students remain engaged, which in turn affects the effectiveness of their teaching strategies. Along these lines, Brown and Zhao (2021) stated that the complexities of managing a large classroom can overwhelm teachers, making it challenging to implement and sustain effective pedagogical strategies. Their research shows that the increased need for classroom control and discipline can divert teachers' attention away from instructional activities, negatively impacting the quality of instruction and further complicating efforts to maintain an inclusive classroom environment where all students receive appropriate attention and support (Saikia and Ranjan 2025).

Furthermore, Whitley et al. (2019) confirm that teacher beliefs, experience, and class size are strongly associated with their DI practice. Bhandari et al., (2024) add to this by outlining significant implementation challenges: tailoring instruction to students with varying readiness,

cultural norms, and proficiencies; managing diverse learning outcomes; and addressing student resistance. These burdens often lead to teacher burnout, especially when institutional support is insufficient.

The impact of class size also extends to evaluation practices. Hargreaves and Fullan (2021) note that with an increasing number of students, teachers may struggle to track each student's progress and provide timely feedback effectively. This difficulty in assessment can hinder teachers' ability to identify and address learning gaps and adjust instructional strategies accordingly. Similarly, Williams and Thompson (2022) state that large class sizes can lead to less frequent and less detailed assessments due to time constraints and the volume of students. As a result, this limitation can impact the ability to accurately gauge student performance and tailor instruction to meet individual needs. Corroborating these points, Rosadi et al. (2024) found that it is challenging to manage a class with many students in DI implementation, as it is hard to give equal attention to all students. The study also stated that having too many students in a class makes it difficult for teachers to meet diverse students' needs, as each student has a different learning style and level of understanding. This aligns with the view that the more learners there are, the less individualized support and feedback each one receives.

Finally, high enrollment numbers create logistical barriers to collaboration. Lai and Zhao (2021); Howard (2023) state that while small classes facilitate the frequent, meaningful interactions necessary for developing communicative competence, large classes restrict group work, reducing the effectiveness of collaborative activities. In addition, teachers reported that the large class size was the reason they did not implement the method. The researchers suggest that it should be 15–20 people to address differences in the differentiated instruction implementation process, to organize and manage groups, and to ensure the presentation of products. Teachers in

public schools stated that their class sizes are high, and therefore, they cannot implement not only differentiated instruction but also student-centered activities. Ultimately, this can hinder the implementation of effective instruction, as it is hard to address each student's needs (Toprak and Güneş 2019). Onyishi and Sefotho (2020) further highlight that additional difficulties encountered by teachers include the non-availability of teaching and learning materials, limited classroom space to rearrange seats to create learning centers or work stations where students can complete their assignments, rigidity in the school timetable, and linking formative assessment in the planning of instruction.

Inadequate support

According to Shareefa et al. (2019), insufficient administrative and peer support hinders the effective adoption of differentiated instruction. Teachers require guidance, collaboration, and encouragement from school leaders and colleagues to implement strategies successfully. Consequently, lack of support can reduce teachers' motivation and confidence in trying new methods. This barrier highlights the importance of a supportive school environment for sustaining differentiated instruction practices.

In this context, inadequate support refers to insufficient assistance, resources, or guidance provided to individuals or groups, hindering their ability to achieve desired outcomes or cope effectively with challenges. This lack of proper support can occur across contexts such as education, healthcare, the workplace, or social services, often leading to frustration, decreased performance, and a decline in overall well-being. For this reason, addressing inadequate support is essential to fostering growth, ensuring equity, and promoting success in any environment. Support systems such as training, adequate classroom resources (like projectors, computers), and institutional backing are critical but often lacking, especially in rural and under-resourced schools.

Ultimately, this infrastructural deficit restricts teachers' ability to deliver varied instructional strategies that meet diverse student needs (Ab Hajis and Othman 2024).

On the other hand, according to Hofer and Steininger (2025), most teachers support differentiated pedagogy, citing its advantages in satisfying student needs, igniting interest, and achieving learning objectives in chemistry education. Corroborating this view, Am et al. (2023) found that supported teachers who use differentiated instruction strategies achieve better student academic performance, particularly in subjects such as mathematics, by adapting content, processes, and the learning environment to individual readiness and interests. This suggests that institutional support and resources help overcome barriers such as large class sizes and limited planning time, leading to more consistent implementation and higher student engagement. However, the decision to implement differentiated instruction should be based on a careful analysis of students' needs and goals, as well as the resources and support available to the teacher and the school (Rohaeti et al., 2020). When executed properly, differentiated instruction helps instructors better meet the unique needs of each student, thereby enhancing academic achievement and fostering a supportive learning environment in the classroom (Woolcott et al. 2021).

Despite these benefits, Suryati et al. (2023) stated that one key challenge in implementing differentiated instruction was the lack of support from other teachers. Since DI is relatively new, collaboration and sharing experiences are important to build confidence, but not all teachers are open to new approaches. Some were unaware of DI, and others were hesitant to change, making it harder to create supportive environments. Similarly, a lack of support from school administration can demotivate teachers and further hinder DI implementation. In fact, previous DI studies have

shown that teachers face difficulties, including a lack of resources, time, support, knowledge, and training (Lavania and Nor 2020).

Additionally, there is a notable disparity in resources between urban and rural schools, with teachers in less developed areas struggling with outdated materials and insufficient support. Addressing these issues requires a comprehensive approach that includes improving resource allocation, enhancing teacher training, and developing flexible, contextually relevant curricula to better support Filipino language instruction (Santos and Lim 2021). Parallel to these logistical concerns, several challenges were identified, including teachers' level of self-efficacy in using technology to teach, a lack of support and resources for online teaching, and the struggle to motivate and engage students in remote settings (Miralles-Cardona et al., 2021).

Furthermore, Matiyenga and Ajani (2024) reveal that while differentiated instruction is widely endorsed for its potential to address diverse learning styles and backgrounds, its practical implementation in pre-service curricula remains inconsistent. Several barriers emerge, including insufficient training in adaptive teaching methods, limited resources, and inadequate support systems within teacher education institutions. Specifically, Themane (2023) stated that one significant barrier is insufficient training in adaptive teaching methods. This training gap means that novice teachers are often unprepared to meet the diverse needs of their students, leading to less effective classroom practices and diminished student outcomes. Moreover, ongoing professional development is also essential for in-service teachers to maintain and enhance their differentiation skills. As such, professional development programmes should offer a variety of opportunities, such as workshops, coaching, and collaborative learning sessions, specifically focused on differentiated instruction (Bondie et al., 2019).

Moreover, the findings of Indrawati et al. (2024) reveal that the primary barriers to implementing DI include limited teacher training in differentiation strategies, time constraints in lesson planning, and inadequate support systems, such as resources and collaboration opportunities. Reflecting this trend, according to Letzel et al. (2025), teachers' DI understanding was higher than their DI implementation in both public and private schools. Specifically identified possible reasons for the gap between understanding of DI and its implementation included excessive administrative work, limited support from the school, limited school policy on DI implementation, and limited professional development specific to DI.

Heavy workload

Shareefa et al. (2019) explain that a heavy teaching workload is another factor that limits differentiated instruction. Teachers must balance lesson planning, grading, classroom management, and other responsibilities, leaving less time for customizing instruction. The study indicates that workload pressures can lead to burnout and reduce the likelihood of consistently applying differentiated strategies. Consequently, effective implementation requires managing workload or providing additional resources to support teachers.

Broadly defined, a teacher's heavy workload encompasses all the responsibilities and tasks they are expected to complete, often extending beyond classroom teaching. According to Boeskens and Nusche (2021), teacher workload is an international challenge. In Australia, for instance, workload has been identified as a leading factor in poor attraction and retention in the profession (Heffernan et al., 2022). While teaching is considered one of the world's most important and oldest professions (Da Silva et al. 2024), it is common nowadays in many parts of the world for teachers to have a high workload and significant hours (Creagh et al. 2023). Indeed, teachers can spend up to 55 hours per week in the classroom during the term and many hours in

extracurricular activities, such as school coordinator and grade level chairmanship (Stacey et al. 2024).

Furthermore, the study by Jomuad et al. (2021) found that teacher workload is a significant factor influencing the effectiveness of educational practices. Research indicates that high workload can lead to stress and burnout among teachers, impacting their ability to deliver quality instruction and implement innovative teaching methods. Conversely, studies indicate that when teachers experience reasonable workloads, they are better equipped to provide individualized attention and support to students, leading to improved academic achievement and engagement (Kanwal et al., 2023).

In the same vein, reduced workload allows educators to dedicate more time to lesson planning, professional development, and innovative teaching strategies, fostering a more engaging and effective learning environment (Rose and Šika 2019). Furthermore, a lighter workload can mitigate stress and burnout, enhancing teacher well-being and job satisfaction, ultimately contributing to a more motivated, dedicated, and effective teaching force that positively impacts student learning and overall educational quality (Szabó et al. 2022). However, unmanaged teacher workload is a significant impediment to effective instruction, resulting in several key classroom challenges. Overly burdened educators struggle to allocate sufficient time for thorough lesson planning and preparation, potentially compromising the quality and engagement of learning activities (Christogonus-Anyanwu and Imiruaye 2024). Moreover, the combination of teaching duties and the need to implement data analysis using integrated modern technologies, such as electronic boards and projectors, also draws teachers away from direct student engagement and curriculum enhancement (Inegbedion et al. 2020).

Moreover, the effectiveness of educational systems significantly relies on the intersection of several key factors: manageable teacher workload, successful implementation of differentiated instruction (DI), and adequate infrastructural support (Shoukat Ali 2024). Current research emphasizes the multifaceted pressures educators face, including adapting to diverse student needs, managing increasing administrative tasks, and navigating resource limitations (Marshall et al. 2022). This review synthesizes recent literature to examine how these interconnected elements influence the quality of teaching and learning, primarily drawing on studies conducted in Malaysia, Zambia, and other global contexts.

Specifically, excessive teacher workload presents a substantial impediment to the successful implementation of differentiated instruction. Jomuad et al. (2021) found a significant correlation between heavy workloads and increased burnout among teachers, reducing their capacity to personalize learning experiences effectively. Similarly, Shareefa (2023) highlighted that teachers in multigrade classes face time constraints and difficulties in assessing student learning, hindering their ability to fully implement DI. Additionally, Asman et al. (2023) identified additional teacher duties, compounded by teacher shortages, as major contributors to increased workload, particularly in low-enrollment schools. In response, it is essential to streamline administrative tasks and provide adequate planning time to enable teachers to better integrate DI strategies.

The implementation of differentiated instruction is further complicated by infrastructural challenges, particularly the availability of adequate resources and suitable classroom environments. Shoukat (2024) found a positive correlation between physical infrastructure and learning resources with improved teacher performance, underscoring that infrastructural deficits can undermine even the most dedicated teachers' efforts to implement DI. This is supported by Kaziya (2025), who notes that resource shortages remain a significant barrier to DI in high school

mathematics classrooms. Additionally, large class sizes, as highlighted by teachers in Kalomo District, Zambia, make it challenging to implement individual instruction and differentiated strategies effectively (Kaziya 2025). Therefore, providing adequate learning materials, technology, and classroom space is crucial for supporting DI.

Finally, Baraily and Sherpa (2024) further emphasize that excessive teacher workload and infrastructural limitations significantly impede the implementation of differentiated instruction, particularly in resource-constrained environments. Their study highlights that teachers struggle to balance the diverse needs of students with the demands of content modification, lesson planning, and assessment, while also contending with large class sizes and limited access to essential resources. This ultimately hinders their ability to provide the individualized attention and tailored teaching methods required for effective differentiation. To conclude, the interaction between teacher workload, infrastructural limitations, and teacher preparedness creates a complex web of challenges that impact the quality of education. Mengistie (2020) found that while primary school teachers generally understand DI, they often lack specific strategies to manage mixed-ability classrooms effectively. The study also revealed that teachers' limited DI practice is due to a lack of knowledge, time constraints, and resource shortages (Mengistie 2020). To address these interconnected challenges, systemic solutions are needed that include professional development, infrastructural improvements, and workload redistribution, which would empower teachers to effectively meet diverse student needs (Ab Hajis and Othman 2024).

Demographic Profile

Beyond the physical limitations of school facilities and the necessity of administrative support, the successful execution of tailored pedagogy often rests upon the individual characteristics of the educator. While recognizing the importance of teacher collaboration for the

implementation of Differentiated Instruction (DI), other internal factors must be considered. In line with Knauder and Koschmieder (2019), Smit and Humpert (2012), and Letzel (2021), teacher-related variables specifically self-efficacy, attitude, age, gender, and teaching experience are frequently reported as critical determinants of instructional success.

An educator's demographic profile can significantly influence their pedagogical competencies and approach to classroom diversity. For instance, age often dictates the specific strategies a teacher employs, with distinct variations observed between early-career and veteran educators (Pozas et al., 2020). Similarly, gender-based variations have been observed; some studies indicate that female educators may exhibit higher rates of DI implementation or distinct interaction patterns compared to their male counterparts (Papanthymou & Darra, 2022).

Moreover, total years of teaching experience often correlate with increased adaptability and a broader repertoire of instructional strategies (Ginja & Chen, 2020). These variables suggest that a teacher's background fundamentally shapes their role in differentiated instruction and warrants deeper investigation. A teacher's role is not just to deliver content, but act as an adaptive decision-maker whose demographic profile influences those decisions.

Age

Teacher age refers to an educator's chronological age and is often correlated with teaching experience, training, beliefs, and other factors that may influence instructional practices. According to the literature, several factors influence teachers' attitudes towards classroom heterogeneity, including professional training, age, gender, and students' abilities (Schwab and Alnahdi 2020).

Specifically, the study of Papanthymou and Darra (2022) primarily focuses on teachers in the Dodecanese region of Greece. It acknowledges the existing literature indicating differences in DI implementation by teacher age, with this document's findings suggesting that older teachers use specific DI strategies more often than younger teachers, thereby contrasting with some studies that found a negative correlation between age and DI use. Similarly, differences are also found in teachers' knowledge of differentiated instructional practices as a function of age, teaching experience, and qualifications (Moutlas, 2021; Bellou, 2019).

However, other research presents a more nuanced view; for instance, Pozas and Letzel-Alt (2023) found that teacher age, along with other sociodemographic characteristics, did not have a consistent impact on DI implementation. This suggests that factors like self-efficacy, attitudes towards inclusion, and specific forms of collaboration (synchronization and co-construction) play more significant roles in shaping DI practices, often overshadowing any direct effects of teacher age. In a different light, Moutlas (2021) found positive correlations between understanding of differentiated instruction and education and between teaching experience and age, while negative correlations were found between the use of differentiated instruction and age and teaching experience.

Taking these variables into account, Knauder and Koschmieder (2019) and Letzel (2021) recognize the importance of teacher collaboration for the implementation of DI; other factors to be considered include teacher-related variables such as teachers' self-efficacy, attitudes, age, gender, and teaching experience, which are frequently reported. As Melesse and Mekonnen (2020) note, teacher educators' contributions to prospective teachers' development of multicultural competence are essential, and teachers' training and attitudes, which may vary with age, play a crucial role in their ability to implement DI effectively. Furthermore, Pozas and Letzel-

Alt (2023) demonstrated distinct teacher profiles based on attitudes towards DI, suggesting that varying experiences across age groups might influence approaches to DI. Supporting this notion, the study by Lindner et al., (2021), which focused on classroom settings, highlights how teaching practices vary across educational environments and are influenced by factors that may correlate with teacher age.

Consequently, teacher age as a demographic factor can influence the adoption and implementation of DI. Additionally, Pattier (2021) stated that younger, newly trained teachers might readily embrace contemporary pedagogical approaches, including technology-integrated DI strategies, regarding the use of YouTube as an educational resource. In contrast, Schwab and H. Alnahdi (2020) emphasized that experienced teachers bring a wealth of classroom management skills and a deep understanding of diverse student needs, which are crucial for effective DI. Research suggests that teachers' experience is positively related to their self-efficacy in implementing inclusive practices. However, these experienced teachers might face challenges in adapting to new technologies or shifting from traditional methods to more student-centered DI approaches (Gheyssens et al. 2020). Ultimately, these findings underscore the importance of ongoing professional development for teachers to effectively implement DI, regardless of their experience level or age, as supportive interventions can mitigate any negative impact of age-related differences in DI practices (Gheyssens et al., 2020).

Sex

Teacher sex is a complex variable with potentially subtle influences on teaching practices. While research on its direct impact on differentiated instruction is still emerging, studies suggest that gender may interact with factors such as teacher attitudes and the classroom environment to influence pedagogical choices (Pozas et al., 2020). Understanding these nuances is crucial for

creating equitable and effective learning environments that cater to diverse student needs. For instance, Alharbi and Alsehli (2025) stated that female teachers exhibit more positive teaching attitudes and greater engagement with DI, which correlates with improved student outcomes; however, this conclusion requires further validation across diverse contexts. This underscores the need to examine how gender intersects with other factors that impact differentiated instruction.

In support of this trend, the study by Gülay and Altun (2022) found that female primary school teachers' perceptions of applying differentiated instruction were significantly higher than those of male primary school teachers. In addition, a significant difference was found in all sub-dimensions in this direction. However, contrary to these findings, Saif et al. (2024) found no significant difference in male and female teachers' opinions on DI's value in supporting students with disabilities, challenging the assumption that gender inherently influences DI perception. This suggests that factors like training and experience may be more salient. Nevertheless, other studies indicate potential gender-based differences in the selection of instructional strategies and in the cultivation of a supportive classroom environment, both crucial aspects of Differentiated Instruction (Pozas et al. 2020). Further reinforcing the idea of gender neutrality in certain programs, Gusti et al. (2025) investigation of the Guru Penggerak program in Indonesia found minimal correlation between teacher gender and satisfaction or implementation of learned techniques, suggesting the program's effectiveness is consistent across genders. Similarly, Widayati, MacCallum, and Woods-McConney (2021) found no significant impact of gender on the adoption of professional learning methods, provided educators had equal access to training.

Consequently, this suggests that well-designed professional development can mitigate potential gender-based disparities in implementing new teaching methods, including a gap in teacher education programs, which often fail to adequately prepare future teachers for

sustainable gender-sensitive pedagogies, underscoring the necessity of integrating gender mainstreaming into curricula to promote equitable DI. As a result, gender-responsive DI, inclusive classroom climates, and professional development opportunities should be prioritized to equip all teachers with the competencies to address diverse learning needs equitably, regardless of gender (Miralles-Cardona et al. 2021).

Beyond pedagogical preferences, systemic inequalities also play a role; Punzalan et al. (2025) note that resource scarcity is more common among women. There is also a lack of training, which could influence the point: if those teachers are women and are struggling with this, then it should be a bigger deal to the government. However, those who make the program do not often reflect on this, which is why there is a lot of stress and burden on the teacher, leading to frustration. Furthermore, the study by Galián et al. (2023) also highlights differences in how male and female teachers carry out their educational work, with women being more involved and thus transmitting gender roles, a phenomenon observed in contexts such as Colombia.

Years of Teaching

Years of teaching experience refer to the time a teacher has spent actively instructing students, typically in formal settings. It encompasses planning, delivering instruction, and evaluating learning. In this regard, years of teaching experience and contextual educational experience are significant predictors of a teacher's ability to adopt effective teaching practices (Podolsky et al., 2019). As a practical example, nine Indonesian EFL GP teachers teaching in secondary schools participated in a study where we recruited six GPs—five female teachers and one male teacher. They were 33–47 and had 11–18 years of teaching experience. GP teachers were selected for this study based on their enrolment in DI training as part of the GP program, which guaranteed they possessed the required knowledge. Providing these teachers with a thorough

understanding of DI was expected to enable them to successfully incorporate it into their classrooms and showcase creativity in their teaching (Ayuningtyas et al., 2023).

Building on this, Langelaan et al. (2024) highlights practical classroom experience as the most advantageous factor in developing DI skills, noting the correlation between teacher self-efficacy and years of teaching. Similarly, Van Geel et al. (2022) identifies "gaining experience" and "attitudes and beliefs" as the most crucial elements in developing DI competencies. Furthermore, Adeniran et al. (2025) stress that years of teaching provide essential experiential knowledge and confidence that enhance DI implementation. Supporting this view, studies also show experienced teachers employ a broader range of practices to optimize DI (Ginja and Chen 2020).

However, experience alone is often insufficient, as teacher capacity building through ongoing, comprehensive Professional Development (PD) programs is vital for refining DI skills (Panela, 2025). Differences between participants' views and practices based on years of teaching experience may also warrant examination, as previous research on this question has yielded contradictory results. For instance, some studies suggest that experienced teachers are less positive about DI than their novice colleagues (Whitley et al. 2019).

Furthermore, Onyishi and Sefotho (2020) note gaps in empirical research on the relationship between teaching experience, gender, and DI implementation challenges, especially in Nigeria. Many teachers, regardless of experience, face difficulties due to insufficient training and institutional support. In fact, Massey (2025) found that even teachers with two or more years of experience require continuous Professional Development and leadership backing to enhance DI efficacy. Sharp et al. (2025) further highlight that the lack of PD and school leadership support leaves teachers feeling unprepared to differentiate effectively.

Along the same lines, Cantrell et al. (2020) and Kahmann et al. (2022) similarly emphasize the critical role of PD in successful DI application. The pressure to use DI was not anticipated as a

theme before the analysis; however, the interviewees' responses indicate that this is an additional issue that needs to be addressed, especially for teachers who have been in the profession for many years and may rightly feel resistant to external pressures for change. This aligns with earlier findings that teachers' years of teaching experience may be negatively associated with their beliefs in DI (Whitley et al. 2019).

To illustrate the diversity of the teaching force, a study of 67 teacher educators revealed that 38.8% were beginners with 1-3 years of teaching experience, while 32.8% were experienced teachers with over 11 years of teaching experience. MA degree holders accounted for 49.3%, BA degree holders for 31.3%, and PhD degree holders for 19.4%. Colleges and universities provided in-service teacher training on DI, despite a high proportion of untrained teachers (46.3%). The study also included 20 teachers from language studies and pedagogical/educational science, as well as others from social science and natural science departments (Ginja and Chen 2020).

Ultimately, through school-based training, teaching seminars, and experience exchanges, teachers' adaptability and innovation can be continuously enhanced in actual teaching. Improving teachers' professional qualities has become a fundamental support for the implementation of differentiated instruction and a key means of promoting educational improvement (Li 2025). Furthermore, while they acknowledge the rationale for DI and appreciate the benefits it may offer students, methodological uncertainties, implementation difficulties, and a sense of pressure to differentiate—especially for teachers with more years of experience—result in an ambivalent attitude towards the concept.

To sum up, the literature review indicates that several interconnected variables influence the implementation of differentiated instruction (DI) in public schools. Adequate classrooms, electricity, and ICT facilities are also recent infrastructure issues that significantly affect the provision of Differentiated instruction by teachers. Poor classrooms lead to overcrowding and

management problems, whereas poor electricity deprives students of the ability to use technology to learn individually. Poor access to ICT also worsens these issues, as teachers lack access to digital resources and platforms that support the needs of different learners. Some of the implementation issues that directly affect DI are also outlined in the study, including time constraints, limited resources, ignorance, large classes, insufficient support, and cumbersome workloads. Teachers are in a dilemma when it comes to lesson planning and preparation, which is also very crucial to DI. Limited resources, such as teaching materials and equipment, further hinder their ability to teach. The deficit in knowledge of DI strategies and best practices, combined with the high number of students in the classroom, means teachers are unable to offer one-on-one support. Poor school administration and excessive workloads are also factors that lead to teacher burnout and poor-quality instruction. In addition, the paper elucidates that content, process, and product play a crucial role in DI. It is essential to differentiate instruction by varying the learning process to suit the needs of all students, adjusting the content they learn, and allowing students to showcase their learning through a variety of products, making this a practical element of DI. The infrastructural and implementation challenges mentioned above, however, impair these elements. Age and gender of the teacher are also demographic factors that influence the implementation of Differentiated Instruction, but their effects are complex and require further investigation. These thorny issues should be addressed to establish effective and equitable learning conditions in government school systems and thereby enhance student achievement.

1.3 Theoretical Framework

This research on the infrastructural and differentiated instruction implementation challenges in selected public schools is guided by two primary theoretical frameworks: School Building Condition Theory by Mark Schneider (2002) and Instructional Theory by Robert Gagne (1965).

Mark Schneider (2002). School Building Condition Theory

The School Building Condition Theory holds that the school's physical facilities (classroom space, ventilation, lighting, and maintenance) have a substantial effect on teaching and student performance. Schneider noted that well-maintained, adequate, and safe school facilities create a favourable environment for learning and effective teaching. On the other hand, quality instruction is impeded by poor building conditions, congested classrooms, and a lack of instructional facilities. This Theory points out that infrastructure is a significant factor that facilitates or restricts teachers' capacity to implement instructional approaches such as differentiated instruction in the classroom.

Robert Gagne's Instructional Theory (1965)

Instructional Theory focuses on how instructor model, plan, and execute instructional methods to achieve effective learning outcomes. Gagne stated that quality teaching entails establishing learners' needs, defining clear goals, and using appropriate pedagogical approaches and resources to ensure that every student is successful. Implementing differentiated instruction involves teachers altering the content, procedures, and learning activities to meet the abilities, interests, and learning styles of different learners. The importance of Differentiated Instruction relies on the teacher's capacity to organise lessons so that they are meaningful and enable every learner to make progress. The Theory, therefore, provides the basis for understanding how teachers implement instructional practices to accommodate diverse students.

Collectively, these theories have provided a reasonable basis for the study, "Infrastructural and Differentiated Instruction Implementation Challenges of Public Schools." The Instructional Theory describes how teachers plan and use differentiated instruction to meet the learning needs

of students with varying needs. The School Building Condition Theory emphasises the importance of strong infrastructure in facilitating these instructional practices. It demonstrates that teachers depend on it to implement differentiated instruction, underscoring the need for supportive school facilities to enable effective, inclusive teaching.

1.4 Schematic Diagram

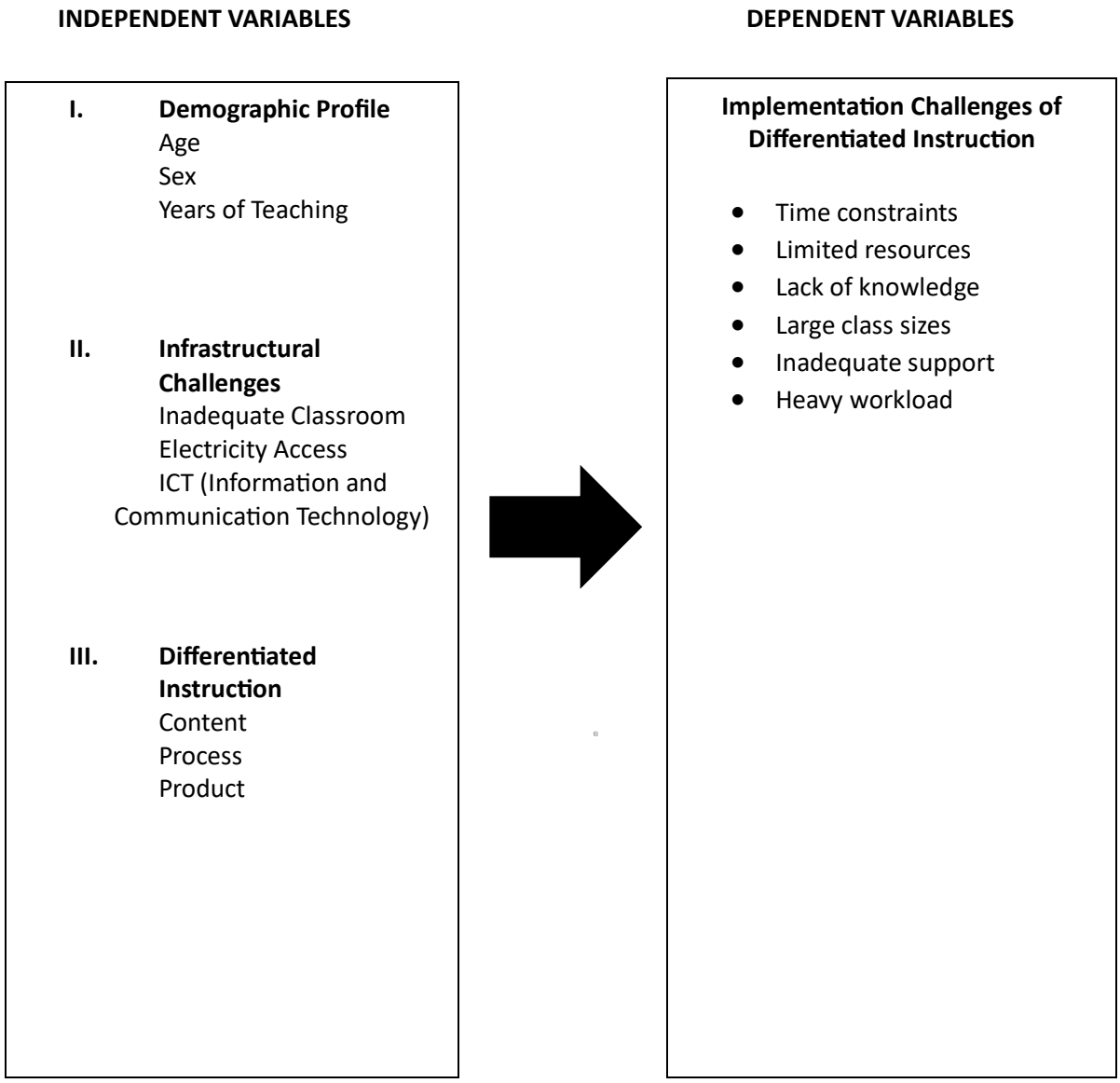


Figure 1. The figure shows the interplay between the independent and dependent variables.

1.5 Statement of the Problem/ Objectives of the Study

This study will examine the infrastructural and differentiated instruction implementation challenges of public-school teachers in Nasipit, Agusan del Norte

Specifically, it will seek to answer the following questions:

1. What is the demographic profile of the teachers in the selected public school in Nasipit, Agusan del Norte, in terms of:
   1.1 Age;
   1.2 sex; and
   1.3 years of teaching?
2. What is the level of infrastructural challenges faced by the teachers in selected public schools in Nasipit, Agusan del Norte, in terms of:
   2.1 inadequate classroom;
   2.2 electricity access; and
   2.3 ICT access?
3. What is the extent of implementation of differentiated instruction by teachers in selected public schools in Nasipit, Agusan del Norte, in terms of:
   3.1 content;
   3.2 process; and
   3.3 product?
4. What is the level of challenges were experienced by the respondents in the implementation of differentiated instructions, in terms of:
   4.1 time constraints;
   4.2 limited resources;
   4.3 lack of knowledge;

4.4 large class;

4.5 inadequate support; and

4.6 heavy workload?

5. Is there a significant relationship between infrastructural challenges and implementation challenges of the Differentiated Instruction?

1.6 Hypothesis of the Study

The following hypothesis guided by the study and tested at a 0.05 level of significance.

HO_1: There is no significant relationship between infrastructural challenges and implementation challenges of differentiated Instruction at Selected Public School in Nasipit, Agusan del Norte.

1.7 Scope and Limitations

This study is designed to explore and examine the challenges of implementing infrastructural and differentiated instruction in selected public schools in Nasipit, Agusan del Norte. It specifically focuses on the experiences and perspectives of public- school teachers from Calixto D. Montinola National High School, Culit National High School, and Nasipit National High School.

Respondents will be selected using purposive sampling to ensure fair representation across different fields of specialization. Data collection will be conducted exclusively during School Year (SY) 2025–2026. Consequently, the findings will provide a contextual understanding of the challenges that teachers in the selected schools in Nasipit faced during the specified academic year.

The scope and generalizability of this study are restricted in two key areas. First, the investigation is limited to selected public schools within Nasipit, Agusan del Norte, and therefore, the findings may not be generalized to private schools or public schools in other geographical

areas. Second, the study is strictly confined to infrastructural and differentiated instruction variables, thereby excluding other potential influences on teaching and learning, such as administrative policies, curriculum revisions, parental involvement, and learners' socio-economic conditions.

1.8 Definition of Terms

To ensure a clear understanding of key concepts, the following terms have been defined operationally within the context of this study;

Age. It represents the respondents' chronological age, categorized into ranges (e.g., 20–29, 30–39, 40–49, and 50 and above), in selected public schools in Nasipit, Agusan del Norte. This variable is used to examine potential trends in differentiated instruction (DI) practices across different age groups.

Content. It refers to the lessons, topics, and materials of teachers in a selected public school in Nasipit, Agusan del Norte, to adjust their instruction to meet students' varying needs.

Electricity Access. It indicates the extent to which teachers in selected public schools in Nasipit, Agusan del Norte, have reliable electrical power in their classrooms and school facilities

Inadequate Classroom. It represents the degree to which teachers in selected public schools in Nasipit, Agusan del Norte, are characterized by insufficient space, overcrowding beyond regulated student-teacher ratios, and unsuitable physical conditions.

Heavy workload. It signifies teachers' perceptions in selected public schools in Nasipit, Agusan del Norte, regarding the volume and intensity of the teaching and non-teaching tasks they handle.

ICT (Information and Communication Technology). It pertains to the availability and use of digital technologies by teachers in selected public schools in Nasipit, Agusan del Norte, to support teaching and learning, including computers, internet access, audiovisual tools, and educational software.

Inadequate Support. It presents the teachers in selected public schools in Nasipit, Agusan del Norte, regarding the sufficiency of professional guidance, resources, and institutional support in implementing differentiated instruction.

Lack of knowledge. It captures the teachers in selected public schools in Nasipit, Agusan del Norte, regarding their insufficient understanding, skills, or exposure to differentiated instruction.

Large class sizes. It conveys a high student-to-teacher ratio in classrooms in selected public schools in Nasipit, Agusan del Norte, making it difficult for teachers to provide individualized support and implement differentiated instruction.

Limited resources. It concerns the teachers in selected public schools in Nasipit, Agusan del Norte, regarding the lack of adequate teaching materials, classroom equipment, and school facilities, which hinder their ability to deliver effective instruction.

Process. It is characterized by specific teaching methods and strategies used by teachers in selected public schools in Nasipit, Agusan del Norte, to implement differentiated instruction.

Product. It encompasses the various forms of student output in the classroom in selected public schools in Nasipit, Agusan del Norte, including tests, projects, reports, presentations, and other assessments, which teachers use to evaluate learning outcomes in differentiated instruction.

Sex. It denotes the teachers' self-identified sex in selected public schools in Nasipit, Agusan del Norte. Categorized as male and female, to examine the potential differences in attitudes and practices in the implementation of differentiated instruction.

Time constraints. It describes the teachers in selected public schools in Nasipit, Agusan del Norte, regarding insufficient time to plan and implement differentiated instruction.

Years of Teaching. It specifies the total number of years that teachers in the selected public school in Nasipit, Agusan del Norte, have spent in active instructional service. It is used as an indicator of a teacher's knowledge, experience, and capacity to implement differentiated instruction effectively despite existing infrastructural and instructional challenges.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter presents the research design, locale, respondents, research instruments, ethical standards, data-gathering procedure, and data analysis plan. These components collectively describe the systematic process that guided the conduct of the study entitled "Infrastructural and Differentiated Instruction Implementation Challenges of Selected Public Schools.

2.1 Research Design

This study will employ a descriptive-correlational research design to determine the relationship between the independent variables Infrastructural Challenges and Differentiated Instruction—and the dependent variable, Implementation Challenges in selected public schools in Nasipit, Agusan del Norte. This design is appropriate because it allows the researcher to describe the current conditions of each variable while also examining whether significant relationships exist among them without manipulating any conditions. The independent variables consist of Infrastructure (inadequate classroom facilities, electricity access, and ICT access) and Differentiated Instruction (content, process, and product). The dependent variable is Implementation Challenges, which include time constraints, limited resources, lack of knowledge, large class sizes, inadequate support, and heavy workload. Using a correlational approach, the study will measure how these variables naturally occur and assess the degree to which changes in infrastructural challenges and differentiated instruction are associated with the challenges teachers face. Descriptive statistics will be used to present the levels of each variable, while correlational analysis will determine the strength and significance of relationships among them.

2.2 Research Materials and Methods

The main research instrument will be a researcher-made survey questionnaire. It is composed of four parts: (1) demographic profile of the respondents (age, sex, and years of teaching experience), (2) infrastructural challenges, (3) differentiated instruction, and (4) implementation challenges of selected public schools in Nasipit, Agusan del Norte. All items will use a four-point Likert scale. The questionnaire will be validated by experts and pilot-tested by the researchers to ensure reliability.

2.3 Research Environment, Population, and Sample

The researchers will be conducting the study in three public high schools in Nasipit, Agusan del Norte: Calixto D. Montinola National High School, Culit National High School, and Nasipit National High School. These schools represent diverse educational settings in terms of location, historical development, and infrastructural context.

Calixto D. Montinola National High School, located at E. Felias St., Brgy. 4, is situated in an urban area with moderate access to educational resources. The school traces its roots to Nasipit Integrated School, which opened in June 2000 with 135 students and three teachers under the leadership of Dr. Emelie Canoy. It was later absorbed by Nasipit National Vocational School during the 2001–2002 school year and subsequently operated as the NACES Campus of Nasipit National High School. Through Republic Act No. 10002, signed on February 23, 2010, it officially became Nasipit National High School. Due to persistent overcrowding and unsuccessful consolidation efforts, initiatives for school separation resumed when Cesar V. Balanghig returned as principal in 2023. The school was eventually established as an independent institution and named after Calixto D. Montinola, the land donor who contributed 879.68 square meters for the school site. It was

approved to operate on June 6, 2024, formally launched in July 2024, and has since undergone leadership transitions, with Jonchel M. Fran assuming the principalship in September 2025.

Culit National High School, located at D-3, Culit, Nasipit, Agusan del Norte, is situated in a rural area with relatively limited infrastructural support. Its establishment was initiated through a series of resolutions by the Sangguniang Barangay of Culit, beginning with Resolution No. 54 in July 2003, which authorized coordination with the Schools Division Office for the opening of a barangay high school. The school expanded its academic offerings in 2008 with the approval of additional year levels under Culit Integrated School. Further resolutions, endorsement letters, and a Memorandum of Agreement in 2014 formalized the possession of the school site and ensured the continued operation and institutionalization of Culit National High School.

Nasipit National High School, located at Ata-atahon Central 1, serves a coastal rural community that faces environmental and geographical challenges, including vulnerability to weather-related disruptions affecting infrastructure stability and educational continuity. Its history began in June 2004 when Barangay Ata-Atahon requested the use of a human settlement building for secondary education. The Department of Agrarian Reform transferred the building to the Local Government Unit of Nasipit in October 2005. Issues raised during a public hearing in March 2009 regarding an existing integrated secondary school prompted plans for relocation, leading the LGU-Nasipit to designate a 10,000-square-meter lot for a new high school site. The school was formally established through Republic Act No. 10002, signed on February 23, 2010, and was later operationalized following a municipal resolution in October 2010 requesting funding and implementation support from the Department of Education.

The selection of these schools will be intended to provide a variety of educational environments urban and rural, from inland to coastal each with its own peculiar demographic and

infrastructural features. Such a multi-site method will allow the researchers to focus comprehensively on the variables of interest. It will enhance the depth and generalizability of the research findings by encompassing varied school environments representative of different community profiles.

Research Respondents

The study's respondents are determined using a purposive sampling technique, and there will be three public high school teachers from Nasipit, Agusan del Norte.

The respondents of this study will be the teachers who will handle various subjects in the selected public schools in Nasipit, Agusan del Norte. This study has a total sixty (60) respondents, where twenty-nine (29) teachers in Calixto D. Montilla, fifteen (15) teachers in Culit National High School, and sixteen (16) teachers in Nasipit National High School.

Table 1. The Distribution of Respondents of the Study

	Male		Female		Total	
School	**F**	**%**	**F**	**%**	**F**	**%**
Calixto D. Montinola National High School	8	42.11	21	51.22	29	48.33
Culit National High School	6	31.58	9	21.95	15	25.00
Nasipit National High School	5	26.32	11	26.83	16	26.67
Total	**19**	**31.67**	**41**	**68.33**	**60**	**100**

Table 1 represents the distribution of respondents across different schools, broken down by gender. It shows the number of male and female respondents from each school, along with the total number of respondents per school.

Research Sampling

The researchers will use purposive sampling in selecting the respondents. Specifically, teachers will be selected based on the following criteria: (1) must be currently employed in selected public schools; (2) must be teaching various subjects; (3) must have at least one year of teaching experience; (4) must be directly involved in implementing differentiated instruction; and (5) must be willing to participate in the study. These criteria will ensure alignment with the objectives of the study and the collection of accurate and valid data on infrastructural challenges, differentiated instruction, and implementation challenges.

Research Instruments

The main research instrument used in this study is a researcher-made questionnaire designed to gather accurate and sufficient data about the infrastructural challenges, differentiated instruction, and implementation challenges of public high school teachers. The questionnaire is developed by using validated tools, established literature, and relevant theories on teacher stress to ensure it is valid, aligned with the topic, and reliable

The questionnaire consists of four (4) major sections, with each section representing a main variable of the study: (1) Demographic Profile, (2) Infrastructural Challenges, (3) Differentiated Instruction, and (4) Implementation Challenges of Differentiated Instruction. Each section is further divided into parts, which correspond to the identified sub-variables. The respondents will answer each item using a 4-point Likert scale to measure respondents' level of

agreement or frequency of experience. The scale is described as (4) Strongly Agree (SA), (3) Agree (A), (2) Disagree (D), (1) Strongly Disagree (SD). The questionnaire will undergo content validation and pilot testing by the researchers to ensure reliability.

Section I: Demographic Profile

This section aims to describe the background characteristics of the respondents, to determine the profile of teachers in terms of age, sex, and years of teaching experience. It further aims to examine the relationship between these profile variables and the challenges teachers face in implementing infrastructural and differentiated instruction implementation challenges in selected public schools.

Section II: Infrastructural Challenges

This section examines the infrastructural challenges experienced by public high school teachers, which may affect the quality of instruction and the effective implementation of teaching strategies. The study focuses on the following variables: inadequate classrooms, electricity access, and ICT access, as identified in the study of Navarro (2024). The study's questions regarding these infrastructural challenges are developed based on the review of related literature

Each part contains ten (10) items, for a total of thirty (30). The respondents will answer each item using a 4-point Likert scale:(4) Strongly Agree (SA), (3) Agree (A), (2) Disagree (D), (1) Strongly Disagree (SD).

Part I. Inadequate classroom

This section assesses the classroom environment, including overcrowding, limited furniture, poor ventilation and lighting, noise, and restricted space. It captures how these

conditions affect teaching effectiveness, lesson continuity, student engagement, and learning outcomes.

Part II. Electricity Access

This section assesses the availability and reliability of electricity in schools, capturing how power outages and limited access to electrical and ICT tools affect teaching, student engagement, and lesson delivery.

Part III. ICT Access

This section assesses the availability and use of ICT tools, including computers, internet, and digital teaching resources, and how limitations affect teaching effectiveness, technology integration, and student learning outcomes.

Section III: Differentiated Instruction

This section examines differentiated instruction as experienced by public high school teachers. The study focuses on three key variables: content, process, and product, as identified by Ab Hajis and Othman (2024). The research questions related to differentiated instruction were developed based on a review of relevant literature.

Each part contains ten (10) items, for a total of thirty (30). The respondents will answer each item using a 4-point Likert scale:(4) Strongly Agree (SA), (3) Agree (A), (2) Disagree (D), (1) Strongly Disagree (SD).

Part I: Content

This part assesses how teachers adapt lessons, materials, and resources to match

students' readiness, interests, and learning profiles, and how these adjustments support diverse and engaging learning experiences.

Part II: Process

This part assesses how teachers adapt tasks, grouping, and instructional strategies to match students' readiness, interests, and learning styles, promoting engagement and inclusive learning.

Part III: Product

This part assesses how teachers adapt assessments and learning outputs to students' abilities, interests, and learning styles, promoting engagement, creativity, and personalized learning.

Section IV: Implementation Challenges

This section examines differentiated instruction as experienced by public high school teachers. The study focuses on six key implementation challenges: time constraints, limited resources, lack of knowledge, large class sizes, inadequate support, and heavy workload, as identified by Shareefa et al. (2019). The research questions regarding these challenges were developed based on a review of related literature.

Each part contains ten (10) items, for a total of thirty (30). The respondents will answer each item using a 4-point Likert scale:(4) Strongly Agree (SA), (3) Agree (A), (2) Disagree (D), (1) Strongly Disagree (SD).

Part I: Time Constraints

This section examines how limited time affects teachers' ability to implement

differentiated instruction, including planning, preparing materials, and addressing diverse student needs, and how these limitations influence the consistency and effectiveness of DI.

Part II: Limited Resources

This section examines how limited materials, technology, and support affect teachers' ability to implement differentiated instruction, impacting lesson planning, individualized support, and overall teaching effectiveness.

Part III: Lack of Knowledge

This section examines how teachers' limited knowledge and professional development in differentiated instruction affect their ability to tailor content, processes, and assessments, influencing competence, confidence, and effective implementation.

Part IV: Large Class Sizes

This section examines how large classes limit individualized attention, differentiated instruction, and student engagement, while increasing teacher workload, stress, and challenges in managing diverse learning needs.

Part V: Inadequate Support

This section examines how insufficient administrative and peer support, limited resources, and lack of professional development hinder teachers' ability to implement differentiated instruction effectively, while strong support enhances engagement, teaching quality, and student outcomes.

Part VI: Heavy Workload

This section examines how heavy teaching responsibilities, administrative duties, and large class sizes limit teachers' capacity to plan, customize instruction, and implement

differentiated instruction effectively, affecting teaching quality and student engagement.

2.4 Research Data and Analysis Plan

Data Gathering Procedure

The researchers will follow a systematic procedure to collect data for this study while ensuring adherence to ethical standards, confidentiality, and the comfort of respondents. The respondents will be selected through purposive sampling, consisting of teachers from the selected public schools who have direct experience in implementing differentiated instruction and who teach core subjects aligned with the curriculum.

This study will use a researcher- made survey questionnaire designed to collect quantitative data on demographics, infrastructure, differentiated instruction, and implementation challenges, all measured on a four-point Likert scale. The questionnaire will undergo content validation by an expert and pilot testing to ensure reliability and clarity. All collected data will be treated confidentially and processed anonymously to maintain the privacy of respondents and enhance the credibility of the study.

The researchers will follow the following process to gather the data needed for the study: before data collection, permission will be requested from the dean and the research adviser. Once approval is granted, official permission letters will be obtained from the Principal of Nasipit Public School, Calixto D. Montinola; the Principal of Culit National High School; the Principal of Nasipit National High School; and the researcher's adviser for pilot testing. An orientation of the respondents will be conducted before the distribution of the questionnaire. The group discussion will be conducted with the principal's permission, and the retrieval of the questionnaire will be

clearly explained to the respondents, allowing them to withdraw without facing any consequences.

1. **Letter of Approval**. The researchers will secure the permission from the dean and the research adviser and the school principal in Calixto D. Montinola, Culit National High School, and Nasipit National High School.
2. **Orientation**. The researchers will conduct a group discussion with the respondents to inform them about the study's purpose, procedures, potential risks and benefits, and the ethical agreement involved. The respondents will be assured that their responses will be kept safe, will not be exposed, and will be treated with strict confidentiality.
3. **Distribution of Questionnaires**. The researchers will distribute the survey questionnaire face-to-face, and the respondents will answer the provided questionnaire. The respondents will be given ample time to submit the required data for the study. It will be considered and understood during the respondents' availability and break times, with their coaches' or teachers' consent.
4. **Retrieval of the Questionnaire**. The researchers will retrieve the responses and prepare for data collection.
5. **Gathering of Data**. The data will be collected, checked, tallied, presented, analyzed, and interpreted using statistical tools.

Statistical Treatment

1. **Frequency and Percentage**. It will answer statement problem one on determining the demographic profile of the teachers in the selected public school in Nasipit, Agusan del Norte in terms of age, sex, and years of teaching experience.

2. **Weighted Mean**. It will answer problems two, three, and four. Problem two: What extent do teachers in selected public schools in Nasipit, Agusan del Norte, experience infrastructural challenges in terms of adequate classrooms, electricity access, and ICT access? Problem three: What is the extent of implementation of differentiated instruction by teachers in selected public schools in Nasipit, Agusan del Norte, in terms of content, process, and product? Problem four: What is the level of challenges were experienced by the respondents in the implementation of differentiated instructions, in terms of time constraints, limited resources, lack of knowledge, large class size, inadequate support, and heavy workload?
3. **Correlational Analysis.** This study sought to determine whether a significant relationship exists between infrastructural challenges and implementation challenges of Differentiated Instruction (DI) in selected public schools. The scores obtained from respondents representing the degree of infrastructural limitations and level of implementation difficulties were subjected to correlational analysis using Pearson Product-Moment Correlation.

2.5 Research Ethical Standards

In this research, entitled "Infrastructural and Differentiated Instruction Implementation Challenges of Selected Public Schools," several ethical considerations will be addressed to ensure that the study meets the required standards for research integrity and the protection of respondents.

A. Protection of Intellectual Property Rights (IPR)

The researchers will develop a researcher-made questionnaire to gather accurate and sufficient information on the implementation of infrastructure and differentiated instruction in the selected public schools. The instrument will be in the form of a survey questionnaire and will not involve the use of any new technologies, procedures, or products. The researchers will retain the intellectual property rights to the researcher-made questionnaire, ensuring protection against unauthorized use, reproduction, or distribution, and safeguarding the originality and integrity of the research instrument

B. Informed Consent

Respondents in the current research will be informed of the researchers' goals and assured that their information will be kept confidential. The questionnaires will be personally distributed to facilitate easy communication with the respondents and to enable them to seek clarification. The researchers will obtain institutional consent from the school administration to implement the study, ensuring that school policies will not conflict with the research and that participants' data will remain secure. Furthermore, respondents will receive a clear and easily comprehensible consent form that will outline their rights, including the right to withdraw from the study without penalty. The process of obtaining informed consent may take longer than expected, particularly for teachers in public schools who will participate in the study, as scheduling will need to accommodate their teaching responsibilities and ensure that all respondents are fully informed about the study's purpose and their rights.

C. Data Privacy and Confidentiality

The researchers will ensure that the data will be accessible only to the research team, the statistician, and the research adviser, who will treat the information with the highest degree of

confidentiality. The identities of the respondents will be kept confidential to protect their privacy. Upon data collection, all the hard copies will be safely stored, and digital copies will be secured with limited access. After data analysis and reporting, digital files will be permanently deleted to protect the respondents' privacy and safety.

D. Voluntary Participation and Freedom to Withdraw

The researchers will use the list of respondents as given by the school principal to name and contact the respondents. All respondents will be informed of their rights, including the right to withdraw from the study without facing any consequences. Participation will not be rewarded or certified, as the aim was to obtain valuable data while respecting participants' freedom, comfort, and voluntary engagement.

E. Minimization of Harm and Risk Management

To minimize potential harm or inconvenience to public school teachers and avoid affecting their professional, psychological, or emotional well-being, the researchers will conduct the study during break times or free periods. Appropriate support services will be made available as needed, including access to the school guidance office, assistance from school administrators, and direct support from the researchers for clarification or debriefing. The questionnaire will be distributed within the school premises with the consent of the teachers and school administration, ensuring that participation is voluntary and does not interfere with regular teaching duties.

F. Beneficence and Contribution to Knowledge

This study will provide significant insights into the current conditions of infrastructure challenges, differentiated instruction and the implementation challenges influencing teachers'

instructional practices. Participation in the study will allow public school teachers to voice their experiences and challenges, reflect on their teaching practices, and contribute to evidence-based recommendations for improving school facilities and instructional support. The findings may inform future professional development programs and administrative decision-making that could benefit teachers' working conditions and instructional effectiveness. All respondents will be fully informed of the study's objectives, and their informed consent will be obtained to ensure transparency, confidentiality, and respect for their participation.

G. Justice and Fair Participants Selection

The respondents of this study will be determined using fair inclusion criteria. The respondents will be selected by using a purposive sampling method, ensuring that every teacher has an equal chance of being part of the study. Throughout the process, all respondents will receive equal quality of treatment before, during, and after the study.

A total of sixty (60) teachers will participate for this study. This number will be determined based on the total population of teacher respondents from the three participating schools. Specifically, twenty-nine (29) teaches respondents will be from Calixto D. Montinola National High School. Fithteen (15) teacher respondents will be from Culit National High School, and sixteen (16) teacher respondents will be from Nasipit National High School.

The researchers will obtain informed consent from all respondents for their involvement in surveys. This selection process will be conducted fairly and without discrimination, in line with the study's purpose of understanding the challenges of implementing infrastructural and differentiated instruction.

H. Data Integrity and Accuracy

Despite the measures implemented to ensure data accuracy and reliability, the study may still be subject to certain limitations and potential biases. The findings rely on self-reported data from teachers, which may be influenced by personal perceptions or response bias. The scope of the study is limited to selected public schools in Nasipit, which may affect the generalizability of the results. Additionally, the use of a researcher-made questionnaire, while carefully developed and reviewed, may have inherent limitations. These factors will be considered in the interpretation of the findings.

I. Transparency and Honesty in Reporting

This study does not involve the use of any patented technologies; therefore, no patent clearance is required. The study will utilize copyrighted literature and secondary data, including academic works and other relevant sources. Permission will be obtained where necessary, and all materials will be properly cited and acknowledged to ensure respect for intellectual property. Furthermore, the research findings will be reported truthfully, accurately, and transparently, without fabrication, falsification, or misrepresentation, in adherence to ethical research standards. In addition, the researchers acknowledge the use of AI-assisted tools during certain stages of the research process; however, a detailed explanation of its responsible use will be provided in the section dedicated to Responsible Use of AI.

J. Use of Patented and Copyrighted Materials

This study will ensure that all patented materials, technologies, or processes referenced or utilized are properly acknowledged and used only with the necessary permissions. No patented work will be reproduced, modified, or distributed without authorization, respecting the intellectual property rights of the creators. Additionally, the study's findings will be reported

truthfully and transparently. All data, results, and interpretations will be presented accurately, without fabrication, falsification, or selective reporting. Any potential conflicts of interest will be disclosed, and all sources will be appropriately cited to maintain academic integrity and honesty in research.

K. Ethical Considerations for Animal and Human Trials

This study does not involve experimental treatments, clinical human trials, or the use of animals; therefore, ethical clearance for animal use is not applicable. The respondents are public school teachers from Calixto D. Montinola National High School, Culit National High School, and Nasipit National High School. Although the study involves human participants, all standard ethical practices will be strictly observed, including obtaining informed consent, clearly explaining the purpose, potential risks and benefits of participation, ensuring confidentiality, and emphasizing voluntary participation.

L. Responsible Use of AI and Other Related Technologies

The study will ensure that the language and grammar used in the research are clear, accurate, and well-constructed to facilitate understanding. While tools such as Grammarly will be used for assistance, the focus remains on presenting essential information accurately and transparently. ChatGPT will be used to find ideas that expand the literature review. To use these technologies responsibly, the researchers will review all AI-generated content and ensure it does not introduce bias or errors. Respondents will be informed that the research process and reporting are conducted with integrity and clarity to maintain the study's validity.

M. Ethical Clearance and Institutional Approval

This study will receive ethical approval from the Institutional Research Ethics Committee (REC) of Saint Michael College of Caraga to ensure compliance with all necessary guidelines on data privacy, intellectual property, and the ethical treatment of human participants. The researchers will provide respondents with a consent form to ensure they complete the questionnaire voluntarily.

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CURRICULUM VITAE



NAME: JUNNA MARIE B. NAVARRO
ADDRESS: District –ilang ilang Brgy. 5, Nasipit, Agusan del Norte
EMAIL: junnamarie_navarro@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : July 3, 2005
Height : 5'0
Weight : 48
Civil Status : Single

Educational Attainment

Elementary : Nasipit Central Elementary School
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
TEACHERS FOR MODERN CHANGE	**BPED REPRESENTATIVE**
MICHAELINIAN PERFORMING	**PERFORMER(DANCER)**

CURRICULUM VITAE



NAME: RECHIEL F. JEVILLANO
ADDRESS: P7, Matabao, Buenavista, Agusan del Norte
EMAIL: rechiel_jevillano@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : March 4, 2005
Height : 5'2
Weight :
Civil Status : Single

Educational Attainment

Elementary : Matabao Elementary School
Secondary : Buenavista National Highschool
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: JAN GIVEA SHANIELLE A. MACARAEG
ADDRESS: D-2 Magkono, Talisay, Nasipit, Agusan del Norte
EMAIL: jangiveashanielle_macaraeg@.edu.ph

Personal Information

Sex : Female
Date of birth : May 22, 2005

Height : 4'11
Weight : 40
Civil Status : Single

Educational Attainment

Elementary : Talisay Elementary School SPED Center
Secondary : Saint Michael College of Caraga
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
MICHAELINIAN PERFORMING ARTS GUILD	**PERFORMER(DANCER)**

CURRICULUM VITAE



NAME: JUSTIN DAVE R. MADRINIAN
ADDRESS: D-8, Lo-oc-2, Carmen, Agusan del Norte
EMAIL: JUSTINDAVE_MADRINIAN@smccnasipit.edu.ph

Personal Information

Sex : Male
Date of birth : April 6, 2005
Height : 5'3

Weight : 64
Civil Status : Single

Educational Attainment

Elementary : Carmen Central Elementary School
Secondary : Our Lady Of Carmen Academy of Caraga Inc.
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
MICAHELINIAN PERFORMING ARTS GUILD	**PERFORMER (DANCER)**

CURRICULUM VITAE



NAME: MHYCO HENDRICH D. PALAO
ADDRESS: Brgy. 10, P-5, Buenavista, Agusan del Norte
EMAIL: Mhycohendrich_palao@smccnasipit.edu.ph

Personal Information

Sex : Male
Date of birth : November 9, 2004
Height : 5'6
Weight : 53
Civil Status : Single

Educational Attainment

Elementary : Buenavista Central Elementary School
Secondary : Saint James High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
MICAHELINIAN PERFORMING ARTS GUILD	**PERFORMER (DANCER)**

CURRICULUM VITAE



NAME: AIRA ALNERA SOBRECARAY
ADDRESS: Poblacion-3, P6, Buenavista, Agusan del Norte
EMAIL: AIRAALNERA_SOBRECARAY@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : March 13, 2002
Height : 5'2
Weight : 49
Civil Status : Single

Educational Attainment

Elementary : Buenavista Central Elementary School
Secondary : Saint James High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



NAME: DEXLE CHRISTIAN G. VAPOROSO
ADDRESS: P-5, BRGY 1, Buenavista, Agusan del Norte
EMAIL: dexlechristian_vaporoso@smccnasipit.edu.ph

Personal Information

Sex : Male
Date of birth : October 8, 2004
Height : 5'3
Weight : 52
Civil Status : Single

Educational Attainment

Elementary : Crogon Elementary School
Secondary : Beunavista National High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

APPENDICES

APPENDIX A- LETTER OF APPROVAL

SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

October 08, 2025

JONCHEL M. FRAN
School Principal I
Calixto D. Montinola National High School
E.Felias St., Brgy4, Nasipit, Agusan del Norte

Dear Sir,

Greetings of Peace and Prosperity!

In partial fulfillment of the requirements for the degree, Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently preparing a thesis proposal entitled: **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools"** In connection with this, we respectfully seek your kind permission to conduct our study at your esteemed institution. As part of the initial stage of our research, we are kindly asking for a copy of the list of teachers categorized by their respective majors or fields of specialization for the school year 2025–2026. This information will serve as a vital reference in identifying participants for our study. We assure you that the list and any subsequent data will be treated with strict confidentiality and will be used solely for academic purposes. No personal or sensitive information will be disclosed.

Additionally, we would like to respectfully request the opportunity to meet with the faculty in a scheduled meeting where we can personally present and explain the details and objectives of our study. For further coordination and scheduling, you may reach our research leader at **09539233079**.

Your approval and support in this academic endeavor will greatly aid us in completing our thesis proposal. We look forward to your favorable response to this request. Thank you very much for your consideration and assistance.

Respectfully yours,

JUNNA MARIE B. NAVARRO
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Approved by:

JONCHEL M. FRAN
School Principal

EMAIL: RIID@SMCCNASIPIT.EDU.PH

SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

October 08, 2025

LYDIE GWEN G. OÑATE-CAR
Head Teacher I
Culit,National High School
D-3 Culit, Nasipit, Agusan del Norte

Dear Ma'am,

Greetings of Peace and Prosperity!

In partial fulfillment of the requirements for the degree, Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently preparing a thesis proposal entitled: **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools"** In connection with this, we respectfully seek your kind permission to conduct our study at your esteemed institution. As part of the initial stage of our research, we are kindly asking for a copy of the list of teachers categorized by their respective majors or fields of specialization for the school year 2025–2026. This information will serve as a vital reference in identifying participants for our study. We assure you that the list and any subsequent data will be treated with strict confidentiality and will be used solely for academic purposes. No personal or sensitive information will be disclosed.

Additionally, we would like to respectfully request the opportunity to meet with the faculty in a scheduled meeting where we can personally present and explain the details and objectives of our study. For further coordination and scheduling, you may reach our research leader at **09539233079**.

Your approval and support in this academic endeavor will greatly aid us in completing our thesis proposal. We look forward to your favorable response to this request. Thank you very much for your consideration and assistance.

Respectfully yours,

JUNNA MARIE B. NAVARRO
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Approved by:

LYDIE GWEN G. OÑATE-CAR
Head Teacher I

EMAIL: RIID@SMCCNASIPIT.EDU.PH

SHERJ PHILIPPINES

INNOVATI CENTER

SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

October 08, 2025

RUBEN D. PARPAN, JR., BSC, EdD CAR,LPT
School Head
Nasipit, National High School
Ata-Atahon, Central 1, Nasipit Agusan del Norte

Dear Sir,

Greetings of Peace and Prosperity!

In partial fulfillment of the requirements for the degree, Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently preparing a thesis proposal entitled: **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools"** In connection with this, we respectfully seek your kind permission to conduct our study at your esteemed institution. As part of the initial stage of our research, we are kindly asking for a copy of the list of teachers categorized by their respective majors or fields of specialization for the school year 2025–2026. This information will serve as a vital reference in identifying participants for our study. We assure you that the list and any subsequent data will be treated with strict confidentiality and will be used solely for academic purposes. No personal or sensitive information will be disclosed.

Additionally, we would like to respectfully request the opportunity to meet with the faculty in a scheduled meeting where we can personally present and explain the details and objectives of our study. For further coordination and scheduling, you may reach our research leader at **09539233079**.

Your approval and support in this academic endeavor will greatly aid us in completing our thesis proposal. We look forward to your favorable response to this request. Thank you very much for your consideration and assistance.

Respectfully yours,

JUNNA MARIE B. NAVARRO
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Approved by:

RUBEN D. PARPAN, JR., BSC, EdD CAR, LPT
School Principal I

EMAIL: RIID@SMCCNASIPIT.EDU.PH

SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

December 09, 2025

ARNOLD C. SOLEN
Public Schools District Supervisor (PSDS)
Nasipit West District
Nasipit, Agusan del Norte

Dear Mr. Solen,

Greetings of Peace and Prosperity,

In partial fulfillment of the requirements for the degree Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently conducting a research study entitled **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools."** This study aims to examine the infrastructural conditions and the challenges in implementing differentiated instruction among teachers in selected public schools in Nasipit.

In line with this, we respectfully seek your approval to conduct our research in the following selected public schools in Nasipit, Agusan del Norte: **Calixto D. Montinola National High School, Culit National High School**, and **Nasipit National High School**. The primary respondents of the study will be the teachers from these institutions.

We assure you that all data gathered will be used solely for academic purposes and will be treated with utmost confidentiality and adherence to ethical research standards. Should you grant permission, we will coordinate with the respective school heads to ensure proper scheduling and to avoid disruption of class activities.

We sincerely hope for your favorable consideration of this request. Thank you very much for your time and continued support for educational research initiatives

Respectfully yours,

JUNNA MARIE B. NAVARRO
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Endorsed by:

MELISA C. MALOLOY-ON, EdD-EM
CTE Dean

BEVERLY D. JAMINAL, EdD-ISRM
Vice President for Academic Affairs and Research

Received and Approved by:

ARNOLD C. SOLEN
Public School District Supervisor

EMAIL: RIID@SMCCNASIPIT.EDU.PH
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SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

December 09, 2025

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JEANIE L. MATIVO, PhD. CESO V
Schools Division Superintendent
Division of Agusan del Norte
J.P Rosales Avenue, Butuan City

Dear Dr. Mativo,

Greetings of Peace and Prosperity,

In partial fulfillment of the requirements for the degree Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently conducting a research study entitled **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools."** This study aims to examine the infrastructural conditions and the challenges in implementing differentiated instruction among teachers in selected public schools in Nasipit.

In line with this, we respectfully seek your approval to conduct our research in selected public schools in Nasipit, Agusan del Norte. The primary respondents of this study will be the teachers from these institutions.

We assure you that all data gathered will be used solely for academic purposes and will be treated with utmost confidentiality and adherence to ethical research standards. Should you grant permission, we will coordinate with the respective school heads to ensure proper scheduling and to avoid disruption of class activities.

We sincerely hope for your favorable consideration of this request. Thank you very much for your time and continued support for educational research initiatives

Respectfully yours,

JUNNA MARIE B. NAVARRO No. 09539733074
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Endorsed by:

MELISA C. MALOLOY-ON, EdD-EM
CTE Dean

BEVERLY D. SAMINAL, EdD-ISRM
Vice President for Academic Affairs and Research

Received and Approved by:

JEANIE L. MATIVO, PhD. CESO V 12/12/25
Schools Division Superintendent

EMAIL: RIID@SMCCNASIPIT.EDU.PH
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SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

November 06, 2025

SHELDEN S. ESPINA
School Principal II
Rizal National High School
Rizal, Buenavista, Agusan del Norte

Dear Ma'am,

Greetings of Peace and Prosperity!

In partial fulfillment of the requirements for the degree, Bachelor of Physical Education (BPED) at Saint Michael College of Caraga, we, the third-year students, are currently preparing a thesis proposal entitled: **"Infrastructural and Differentiated Instruction Implementation Challenges of the Selected Public Schools"** In line with this, we would like to request your permission to conduct a pilot testing of our research instrument among teachers handling subjects in your school. This pilot testing aims to assess the clarity, reliability, and validity of our questionnaire before it is used in the actual data collection.

We assure you that all information obtained will be treated with the utmost confidentiality and will be used solely for academic purposes. Participation of the teachers will be voluntary, and the process will be conducted in such a way that it will not disrupt their teaching duties.

We sincerely hope for your kind consideration and favorable response to our request. Should you need further details or clarification, please do not hesitate to contact our group leader at 0953-923-3079.

Thank you very much for your time and support.

Respectfully yours,

JUNNA MARIE B. NAVARRO
Research Leader

Noted by:

NORQUIZA A. DIANAL, LPT
Research Adviser

Approved by:

SHELDEN S. ESPINA
School Principal

EMAIL: RIID@SMCCNASIPIT.EDU.PH

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APPENDIX B- INFORMED CONSENT

SMCC-REC FORM 4
RESEARCHER'S INFORMED CONSENT TO PARTICIPANTS

Instructions:
1. For research that are not under SMCC, you may attach your consent form and skip this part.
2. For SMCC researchers fill out the forms truthfully. Make sure to declare the necessary risks as stipulated in your ethics declaration.

TITLE OF THE STUDY
Infrastructural and Differentiated Instruction Implementation Challenges of Selected Public Schools

NAME/S OF THE RESEARCHER/S:

Junna Marie B. Navarro	Jan Givea Shanielle A. Macaraeg
Dexle Christian G. Vaporoso	Rechiel F. Jevillano
Mhyco Hendrich D. Palao	Aira Alnera Sobrecaray
Justin Dave R. Madrinian	Click here to enter text.

DEPARTMENT:	**COURSE:**
College of Teacher Education	Bachelor of Physical Education

INTRODUCTION
You are being invited to participate in a research study conducted by researcher/researchers above who are affiliated with Saint Michael College of Caraga.

You have been selected as a potential informant because you meet the criteria necessary for inclusion in our study.

Your participation in this study is entirely voluntary. We encourage you to take your time to review the information provided below. Please feel free to ask any questions or seek clarification on any part of the study that you do not fully understand before deciding whether to participate. Additionally, you may choose to discuss your potential involvement in this research with family members or friends.

Should you decide to participate, you will be asked to sign this consent form. A copy of this form will be provided for your records.

In case of concerns about the study, you may contact the Research And Instructional Innovation Deparment through the following:

F. Email: riid@smccnasipit.edu.ph
G. Contact number: 09778064574

Or the researchers of this study:

1. EMAIL: junnamarie_navarro@smccnasipit.edu.ph

2. CONTACT NUMBER: 09539233079	
PURPOSE OF THE STUDY	This study will examine the infrastructural and differentiated instruction implementation challenges of public-school teachers in Nasipit, Agusan del Norte. Specifically, it will seek to answer the following questions: 1. What is the level of infrastructural challenges faced by the teachers in selected public schools in Nasipit, Agusan del Norte? 2. What is the extent of implementation of differentiated instruction by teachers in selected public schools in Nasipit, Agusan del Norte? 3. What is the level of challenges were experienced by the respondents in the implementation of differentiated instruction?
PROCEDURES OF THE STUDY	Please Check below which type of study is being conducted ☒ For Quantitative and Survey Questionnaires: *If you volunteer to participate in this study, you will be asked to participate by answering the survey questionnaire which you can finish in less than 30 minutes.* ☐ For Interviews/ FGD / Case Studies *If you volunteer to participate in this study, you will be asked to participate by answering the questions in the interview guide which may last 30-40 minutes depending on the length of the interview. Moreover, this will be recorded through audio or video.* ☐ For capstones and experimental studies *If you volunteer to participate in this study, you will be asked to participate by allowing the researchers to use specimens and other data needed related to the research.*

POTENTIAL RISKS AND DISCOMFORTS

You may experience some discomfort during the interview/questionnaires due to the sensitive nature of the topics being discussed. If at any point you encounter questions that cause psychological or emotional distress, you have the right to decline to answer without any consequence. Additionally, you may choose to withdraw from the study at any time if you feel unable to continue discussing the requested information. The researchers prioritize your well-being and will ensure that your comfort and welfare remain their highest priority throughout the study.

Potential discomforts are:

1. Slight nervousness when taking tests.
2. Mild stress during classroom participation.

POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY

This study aims to generate valuable insights that may benefit both public and private sector administrators, as well as human resource managers and policy-makers. The findings and discussions derived from this research can provide evidence-based information that may

inform decision-making processes and strategic planning. Additionally, the results of this study may be of particular interest to government agencies such as:

1. Improve your confidence and problem-solving skills in mathematics.
2. Learn faster and understand lessons better through simplified Vedic methods.

CONFIDENTIALITY

All information gathered in this study will be treated with the utmost confidentiality, in accordance with applicable laws and regulations. Any identifiable information collected will be kept strictly confidential, except in cases where disclosure is necessary to protect your rights or welfare. This confidentiality agreement ensures that the researchers can protect your participation information from being accessed by individuals who are not directly involved in the study. When the results of this research are published or presented at conferences, your identity will not be revealed, and no personally identifiable information will be used.

PARTICIPATION AND WITHDRAWAL

Your participation in this study is entirely voluntary. Choosing not to participate will not result in any penalty or loss of benefits to which you are otherwise entitled. By participating in this study, you are not waiving any legal claims, rights, or remedies. You may withdraw from the study at any time without any negative consequences.

RESEARCH PARTICIPANT'S CONSENT

I have read and understood the information provided above. I have had the opportunity to ask questions, and my questions have been answered to my satisfaction. I understand that I will receive a copy of this consent form. I am aware that I can withdraw my consent at any time and discontinue my participation in the study without any penalties or loss of benefits to which I am otherwise entitled.

Name And Signature of The Partcipant

Date Signed:

IF ILLITERATE

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Name And Signature of The Partcipant

Thumb Print

RESEARCHER'S CERTIFICATION

I have explained the details of the research study to the participant, answered all questions posed, and ensured that the participant fully understands the information provided in this document. I believe that the participant has freely given their consent to participate in the study.

NAME AND SIGNATURE OF THE PARTCIPANT
Date Signed:

Confirmation of Participant/s

NAME AND SIGNATURE OF THE PARTCIPANT
Date Signed:

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INFRASTRUCTURAL AND DIFFERENTIATED INSTRUCTION IMPLEMENTATION CHALLENGES OF THE SELECTED PUBLIC SCHOOLS

Dear Participant,

We are pleased to invite you to participate in a research study that aims to examine the infrastructural challenges, differentiated instruction and implementation challenges affecting teachers' instructional implementation in selected public schools. This study seeks to gather valuable insights that will help improve teaching and learning practices in public schools. Please be assured that all information you provide will be treated with the utmost confidentiality and used strictly for research purposes. Your participation is voluntary and greatly appreciated, as your insights will contribute in enhancing teaching and learning in public schools. We sincerely thank you for your time and cooperation. If you have any questions or concerns, please feel free to contact the researchers. We truly value your support in improving the quality of education and appreciate your participation.

Thank you!

THE RESEARCHERS

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Survey Questionnaire

Part l. Profile of the Respondents

Instructions: Please put a check mark (✓) in the box that best represents your answer for each item. Kindly review each question carefully and select the option that applies to you.

Name: _____ Date: _____
(Optional)

Age:
- ☐ 25-34 years old
- ☐ 35-44 years old
- ☐ 45-54 years old
- ☐ 55 years and above

Sex:
- ☐ Male
- ☐ Female

Years of teaching
- ☐ Less than 1 year
- ☐ 1-5 years
- ☐ 6-10 years
- ☐ 11-15 years
- ☐ 16-20 years
- ☐ More than 20 years

Part II. This part includes the Infrastructural Challenges Questionnaire, which covers **Inadequate Classroom, Electricity Access,** and **ICT Access.**

Directions: Put a check mark (✓) in the column that best represents your answer to each statement about infrastructural challenges

Rating Scale	Verbal Interpretation	Verbal Description
4	Strongly Agree (SA)	The respondents have a very high level of agreement with the statement.
3	Agree (A)	The respondents have a high level of agreement with the statement.
2	Disagree (D)	The respondents have a low level of agreement with the statement.
1	Strongly Disagree (SD)	The respondents have a very low level of agreement with the statement.

Inadequate Classroom	4 Strongly Agree (SA)	3 Agree (A)	2 Disagree (D)	1 Strongly Disagree (SD)
1. The lack of sufficient classroom space affects my ability to organize, deliver, and manage lessons effectively. (Kholisah et al., 2024)				
2. The lack of adequate classroom facilities and resources often require me to adjust or rush routine activities such as transitions and dismissals which disrupt lesson continuity and create challenges in sustaining an orderly and well- managed learning environment. (Impuesto et. al,2024)				
3. Large class sizes limit my ability to monitor student needs closely, provide timely feedback, and implement differentiated instructional strategies that				

support varied learning levels. (Vakili et al. 2024)				
4. Poor classroom layouts, inadequate furniture, poor ventilation, make it harder for me to organize and deliver lessons effectively as these conditions limit instructional flow and reduce opportunities for flexible learning arrangements. (Kamara and Dadhai (2020); Peralta et al., 2025)				
5. Overcrowding in the classroom increases my stress as a teacher due to heightened discipline concerns, excessive noise, difficulty sustaining student attention, and reduced opportunities for meaningful learner interaction. (Fatima et al., 2019; Tshangana et al., 2023)				
6. The lack of functional furniture, reliable electricity, and adequate ventilation in classrooms hinders my teaching performance and restricts my ability to implement varied instructional strategies. (Ahmed et al., 2020)				
7. A well-maintained and safe classroom environment helps me feel more comfortable and confident in teaching, contributing to a more stable and productive instructional process. (Seherrie 2023)				
8. Proper lighting, appropriate seating arrangements, and organized classroom spaces allow me to facilitate more effective teaching and support a conducive learning atmosphere. (Nwokedi 2023); Joshi, 2023).				
9. Teaching in a clean, spacious, and well-ventilated classroom enables me to manage student behavior more				

effectively and maintain better focus during instruction. (Afalla & Fabelicio,2020: Gonzales,2024; Cambay et al., 2024)				
10. Inadequate facilities such as poor lighting, unstable chairs, and insufficient learning materials make my teaching more difficult and less efficient, limiting my ability to deliver high-quality instruction (Ibrahim and Aslam 2025; Bresenio 2025)				

Electricity Access	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. Electricity in school improves my ability to create a better learning environment for students, as it supports the consistent use of instructional materials and classroom technologies. (Cabotaje et al., 2021)				
2. Reliable electricity access in our school allows the use of digital tools such as laptops, projectors, and audiovisual materials, which enhance instructional delivery and student engagement. (Dharmaraj, 2025; Cabotaje et al., 2021)				
3. The availability of electricity in school extends teaching and studying hours, enabling more flexible learning schedules and supporting varied instructional needs. (Capule-Navaarro et al., 2020)				
4. Our teaching and learning activities highly depend on electricity to support technological tools and IT-based instruction, especially when integrating digital learning resources. (Sutoyo & Yudhanto 2023)				

5. Electricity availability enables me to implement differentiated instruction using digital platforms and assessments, allowing learners to receive tailored support. (Langelaan,2024; Halkiopoulo&Gkintoni,20204)				
6. Frequent power interruptions in our area negatively affect my teaching productivity and the completion of tasks, often disrupting planned lessons and instructional activities. (Agayon et al., 2020)				
7. The lack of stable electricity in our school limits access to the internet and online learning resources, reducing opportunities for digital and enriched learning experiences. (Padilla Rodriguez 2021; Mhlanga,2020)				
8. The lack of reliable electricity and necessary electrical equipment (such as lighting, televisions, and computers) discourages well-trained teachers from working in schools, affecting instructional quality. (Mamador et al., 2025)				
9. Unequal access to electricity among schools contributes to educational inequity between urban and rural learners, affecting their opportunities to engage in technology-supported instruction. (Moner-Girona et al., 2025; McLaughlin, 2025)				
10. The current electrical infrastructure in my school is inadequate to meet the demands of teaching and learning, limiting the effective use of educational technologies. (Du Plessis& Mestry 2019)				

ICT Access	4 Strongly Agree (SA)	3 Agree (A)	2 Disagree (D)	1 Strongly Disagree (SD)
3. Teacher educators must possess essential competencies to effectively integrate ICT in teaching and manage professional responsibilities, as these skills are critical for facilitating technology-enhanced instruction. (Hanessy et al., 2023)				
4. The integration of ICT in the classroom significantly contributes to developing students' 21st-century skills, including creativity, collaboration, and digital citizenship, which are essential for lifelong learning. (Chen&Lui,2024)				
5. ICT provides teachers and learners with wide access to information through the internet and supports self-directed learning, enhancing both teaching strategies and student engagement. (Sayaf et al., 2022)				
6. The use of multimedia tools, such as simulations, videos, and learning software, enhances the comprehensibility and engagement of lessons, making complex concepts more accessible to learners. (Qaddumi et al., 2021)				
7. The use of ICT tools such as computers, tablets, projectors, and smartphones make the teaching and learning process more effective and engaging, allowing teachers to implement diverse instructional strategies. (Ahmed et al., 2019; Ravishankar & Wase,2024)				

8. Insufficient training and professional development limit teachers' ability and confidence to integrate ICT effectively, reducing their capacity to use technology to support differentiated instruction. (Papadakis et al., 2021; Garba et al., 2020)				
9. Limited technological infrastructure, including unreliable internet and outdated hardware, restricts the effective implementation of ICT in teaching, which affects lesson delivery and learning outcomes. (Ifinedo,2021;Love & Magaji,2025)				
10. Our school has limited ICT tools, and even the available ones are often underutilized in classroom instruction, which hinders the full potential of technology-enhanced learning. (Ifinedi et al., 2020)				
11. Teachers in rural areas often lack access to individual digital devices and reliable internet connection, creating inequalities in teaching quality and student engagement. (Lembani et al., 2020).				
12. Unstable or limited internet connectivity due to geographical and socio-economic factors hinders effective ICT integration in teaching, restricting access to online learning resources and collaborative tools. (Danao et al., 2024)				

Part III. Talks about the components of differentiated instruction questionnaires which are content, process and product that are being implemented by teachers in public schools in Nasipit, Agusan del Norte.

Directions: Please put a check mark (✓) in the column that corresponds to your answer for each item below about the components of differentiated instruction.

Content	4 Strongly Agree (SA)	3 Agree (A)	2 Disagree (D)	1 Strongly Disagree (SD)
1. I adjust the learning content to match each student's learning profile, considering their readiness, interests, and preferred learning styles, in order to support personalized and effective learning experiences. (Irdhina, 2021; Celik,2019)				
2. I use a variety of instructional materials, such as textbooks, video clips, PowerPoint presentations, and printed handouts, to differentiate content and cater to diverse student needs. (Kumar Kharka & Kinley, 2024; Suryati et al., 2024)				
3. I incorporate visual, auditory, and kinesthetic materials and activities to address diverse learning styles, ensuring that all learners can engage meaningfully with the content. (Kristani, 2024; Hasnawati and Netti 2022)				
4. I modify content complexity and difficulty levels to accommodate the varying abilities of students, providing appropriate challenges for each learner. (Karim et al., 2025; Roiha & Polso,2021)				
5. I form small groups or pairs to facilitate peer learning and active participation, allowing students to collaborate and engage more effectively with the content. (Peralta et al., 2025)				
6. I provide assignments or tasks with varying levels of challenge based on students' readiness and learning needs, ensuring differentiated opportunities for				

learning and skill development. (Kumar Kharka & Kinley, 2024; Karim et al., 2025)				
7. I adapt my teaching methods and content to align with classroom conditions and students' real-life experiences, making learning more relevant and meaningful. (Celik,2019; Del Mar, 2024)				
8. I offer additional support or re-teaching sessions for students who have difficulty understanding the content, helping them achieve the intended learning outcomes. (Suryati et al., 2024; Kumar Kharka & Kinley, 2024)				
9. Time limitations and lack of diverse resources affect my ability to differentiate content effectively, often restricting the implementation of personalized learning strategies. (Onyishi &Sefotho, 2020; Tajik et al., 2024, Yuen et al.,2023)				
10. I incorporate students' interests into the content and instructional materials to increase engagement, motivation, and active participation in learning activities. (Irdhina, 2021; Karim et al., 2025)				

Process	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. I adjust my teaching methods and process strategies to meet the diverse learning needs of my students, ensuring that instruction is responsive and				

inclusive. (Del Mar,2024; Smale Jacobs et al.,2019)				
2. I organize students into small groups, pairs, or collaborative teams based on their needs, interests, and abilities to support more effective learning processes. (Suryati et al., 2024; Kharka & Kinley,2024)				
3. I integrate audio, visual, and audiovisual materials into lessons to support diverse learning processes and accommodate multiple learning preferences in the classroom. (Kharka & Kinley,2024; Suryati et al., 2024)				
4. I monitor students' progress throughout lessons to adjust instruction and provide additional support, ensuring that learning objectives are met effectively. (Kristanti,2024; Frontiers in Education, 2021)				
5. I assign different tasks or projects to students according to their learning processes, readiness, and abilities, promoting personalized and meaningful learning experiences. (Puzio et al., 2020)				
6. I provide varied learning materials and resources to accommodate students' diverse preferences and needs, enhancing engagement and understanding. (Puzio et al., 2020; Qorib, 2024)				
7. I design lessons and activities that consider students' interests, abilities, and prior knowledge to optimize their learning processes and outcomes. (Kristanti,2024; Del Mar, 2024)				

8. Time constraints and the effort required for preparation affect my ability to implement differentiated learning processes effectively, sometimes limiting instructional flexibility. (Suryati et al., 2023; Noor & Golzar, 2024)				
9. Insufficient training, professional development, and peer support hinder my ability to apply differentiated instruction processes, reducing the effectiveness of personalized learning strategies. (Gibbs, 2023; Noor & Golzar,2024)				
10. When I implement flexible and adaptive learning processes, students demonstrate higher engagement, participation, and motivation, leading to improved learning outcomes. (Krishan & Al-rsa'i, 2023; Bagus, 2025)				

Product	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. I use different types of assessments, such as quizzes, projects, presentations, and essays, to allow students to demonstrate their learning in diverse ways that reflect their strengths and understanding. (Del Mar, 2023; Saika, 2025; Ginja & Chen,2020)				
2. I provide students with multiple options for presenting their learning, including interactive presentations, videos, and practical demonstrations, to accommodate different learning preferences and skills.				

(Argyropoulou&Zarifir, 2021; Solihah et al.,2023)				
3. I adapt learning products according to each student's abilities, learning preferences, and strengths, ensuring that assessments are personalized and meaningful. (Ginja & Chen,2020; Kristanti, 2024)				
4. I empower students to choose how they will complete and present their tasks, either individually or in groups, promoting autonomy and active engagement in the learning process. (Helmi et al., 2023; Suryati et al., 2024)				
5. I design learning products that align with students' interests to increase engagement, motivation, and investment in their own learning outcomes. (Handiyani et al., 2021; Hadi et al., 2022)				
6. I use differentiated products to assess both students' understanding of the material and their ability to apply knowledge in meaningful ways. (Suryati et al.,2024; Saika, 2025)				
7. I allow students to demonstrate learning through varied formats, including written, oral, artistic, or performance-based outputs, to cater to different learning strengths. (Kharka & Kinley,2024; Kristanti,2024)				
8. I encourage students to express creativity and critical thinking in the products they create, supporting higher-order thinking skills and innovation. (Voyager,2023; Handiyani et al., 2021)				
9. Limited resources, large class sizes, and insufficient professional development affect my ability to implement differentiated products effectively, sometimes limiting personalized				

assessment practices. (Sharda Massey,2025; Shareefa,2021)				
10. I integrate differentiation strategies, such as tiered assignments, choice boards, and flexible grouping, to ensure all students can demonstrate their learning effectively and according to their individual needs. (Sabir et al., 2024; Hatmanto at al., 2023)				

Part IV. This part includes the implementation challenges Questionnaire, which covers **Time constraints, Limited resources, Lack of knowledge, Large class size, Inadequate support and heavy workload.**

Directions: Put a check mark (✓) in the column that best represents your answer to each statement about implementation challenges.

Time Constraints	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. I I find that preparing lessons for differentiated instruction requires more time than traditional teaching methods, due to the need to plan for varied learning needs and instructional strategies. (Jose et al., 2021; Rosadi et al., 2024)				
2. The time needed to plan and design differentiated lessons often conflicts with other teaching responsibilities, creating challenges in balancing multiple instructional demands. (Kamarulzaman, 2024; Creagh et al., 2023)				
3. Limited instructional time during class hours makes it difficult to implement differentiated instruction effectively, reducing opportunities for personalized learning. (Gibbs, 2022; Unal et al., 2022)				

4. Time constraints affect my ability to provide individualized support to students during lessons, limiting targeted interventions and feedback. (Ybyrayeva & Ziyabayeva, 2023; Rosadi et al.,2024)				
5. Managing differentiated instruction alongside mandated curriculum requirements is challenging due to limited time, making it difficult to address the diverse needs of all learners. (Grecu,2022; Swanson et al., 2021)				
6. Preparing differentiated teaching materials that align with students' interests, abilities, and learning styles is time-consuming, requiring careful planning and adaptation. (Suryati et al., 2023; Rosadi et al.,2024)				
7. Continuous professional development, grading, and other teaching tasks reduce the time available to plan and implement differentiated instruction effectively. (Kamarulzaman,2024; Shah et al., 2020)				
8. Time limitations hinder my ability to integrate ICT and other innovative tools effectively in differentiated instruction, restricting the use of technology-enhanced strategies. (Shah et al., 2020; Estaiteyeh &DeCoito,2023)				
9. The workload and work intensification create "time poverty," limiting the type and extent of differentiated instruction I am able to implement in my classroom. (Creagh et al., 2023; Marshall,2022)				
10. Lack of sufficient planning time prevents me from fully addressing the diverse learning needs of my students, reducing the effectiveness of differentiated teaching strategies. (House et al.,2020; Gibbs&Mckay,2021:Zeirwald et al., 2022)				

Limited Resources	4 Strongly Agree (SA)	3 Agree (A)	2 Disagree (D)	1 Strongly Disagree (SD)
1. Limited availability of teaching materials and classroom resources hinders my ability to implement differentiated instruction effectively, restricting opportunities for personalized learning. (Gibbs, 2023; Edsurge,2019)				
2. Lack of access to digital technologies, flexible learning materials, and instructional tools reduces my capacity to deliver differentiated lessons that address the diverse needs of students. (Doan,2025; Grecu,2023)				
3. Insufficient funding and budget constraints make it difficult to acquire the resources necessary to implement differentiated instruction, limiting the range and quality of instructional materials. (Insorio,2024; Guardian,2024)				
4. Due to limited school facilities and teaching resources, it is challenging to provide individualized support and small-group instruction, which are essential components of differentiated teaching. (Maphumulo & Biccard,2024; Vakili et al., 2024)				
5. Inadequate professional development opportunities hinder my ability to utilize available resources effectively for differentiated instruction, reducing instructional efficiency and innovation. (Prudente et al., 2024; Hatmanto & Rahmawati, 2023)				
6. Poor school infrastructure, such as unsafe or deteriorating facilities, negatively affects my capacity to implement differentiated teaching strategies and maintain an effective learning				

environment. (Wilson & Schulke,2025; Ab Hajis & Othman,2024)				
7. Limited access to curriculum-aligned instructional materials forces me to improvise teaching content, which may compromise instructional quality and learning outcomes. (Steiner,2021;Carrete-Marín et al.,2024)				
8. Resource shortages, including insufficient instructional aids, supplementary materials, and classroom support, impact the overall effectiveness of my differentiated instruction practices. (Gibbs,2023: Maphumulo & Biccard,2024)				
9. Administrative support and resource allocation are often insufficient to consistently implement differentiated instruction in my classroom, affecting both planning and execution. (Hatmanto & Rahmawati, 2023;Ab Hajis & Oathman,2024)				
10. Spatial inequalities and limited access to educational resources across the school system affect my ability to apply differentiated instruction equitably, creating disparities in student learning opportunities. (Navarro,2022; Insorio,2024)				

Lack of Knowledge	**4 Strongly Agree (SA)**	**3 Agree(A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
3. I have limited understanding of the concepts and principles of differentiated instruction, which affects my ability to apply it effectively in classroom teaching. (Yavuz,202; Shareefa et al., 2023)				

4. I find it difficult to identify and address students' individual learning needs due to my limited knowledge of differentiated teaching strategies. (Tiara Nove Ria, 2023)				
5. My ack of training and exposure to differentiated instruction limits my confidence in implementing it effectively with diverse learners. (Sethupathy,2020; Nketsia et al.,2024)				
6. I am not fully aware of when and how to select appropriate instructional approaches to meet the diverse needs of students in my classroom. (Smets,2019)				
7. I struggle to integrate knowledge about differentiated instruction into my existing teaching practices, reducing the effectiveness of personalized learning strategies. (Van Geel et al., 2022).				
8. I believe that insufficient professional development opportunities have hindered my understanding of differentiated instruction and its practical application. (Benjamin & Panesar-Agilar, 2020; Unal et al., 2022)				
9. I lack knowledge of how to design and manage a mixed-ability classroom effectively, which impacts the delivery of differentiated instruction. (Alsamiri,2023; Magableh & Abdullah,2020)				
10. My limited knowledge about students' strengths, weaknesses, and learning styles affects my ability to plan and implement differentiated lessons effectively.				

(Saika & Ranjan, 2025; Hasanah et al.,,2022)				
11. I find it challenging to connect students' prior knowledge with new learning experiences due to insufficient understanding of differentiation techniques. (Hasanuddin et al., 2024; Shalikhah & Nugroho,2023)				
12. I need more institutional and administrative support to enhance my knowledge and skills in differentiated instruction, enabling me to implement it more effectively. (Gibbs,2025; Halse,2022)				

Large Class Size	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. Large class sizes make it difficult for me to provide individualized attention and support to each student, limiting my ability to address their unique learning needs. (Pendi Rosadi, 2024; Chand,2023)				
2. Overcrowded classrooms hinder my ability to monitor students' academic progress effectively and give timely, meaningful feedback. (Hargreaves & Fullan,2021; Williams & Thompson,2023)				
3. Managing student behavior and maintaining classroom order becomes more challenging in large classes, which affects the learning environment. (Smith & Robert, 2023; Brown & Zhao,2021)				
4. It is difficult to organize small-group, collaborative, or student-centered				

activities in an overcrowded classroom due to limited space and resources. (Toprak & Gunes,2019; Lai & Zhao,2021)				
5. Having too many students limits my ability to differentiate lessons and adapt instructional strategies to meet diverse learning styles and abilities (Pendi Rosadi, 2024; Saika & Ranjan, 2025)				
6. The time required to assess and provide feedback to a large number of students makes implementing differentiated instruction less practical. (Hargreaves & Fullan, 2021; Williams & Thompson,2022)				
7. Large class sizes increase my workload, contributing to teacher stress and burnout, which can negatively affect instructional quality. (Bhandari et al., 2025, Brown & Zhao,2021)				
8. Physical limitations, such as inadequate classroom space and fixed seating arrangements, prevent me from using flexible groupings and other DI strategies effectively. (Onyishi & Sefotho,2020)				
9. Due to the high number of students, I find it difficult to ensure that all learners are actively engaged and participating during lessons. (Oteniya,2023; Chand,2023)				
10. I believe that smaller class sizes and improved classroom infrastructure would make implementing differentiated instruction more effective, manageable, and conducive to learning. (Lai & Zhao,2021; Yahya et al.,2024)				

Inadequate Support	4 Strongly Agree (SA)	3 Agree (A)	2 Disagree (D)	1 Strongly Disagree (SD)
1. Lack of administrative and institutional support, such as guidance, supervision, or policy backing, makes it difficult for me to effectively plan and implement differentiated instruction in my classroom (Lavania & Nor,2020; Letzel et al.,2025)				
2. Insufficient training, professional development, and capacity-building programs limit my ability to apply differentiated instruction strategies effectively to meet diverse student needs. (Sepadi,2023; Bondie et al.,2019)				
3. Inadequate access to instructional materials, teaching aids, technology, and other classroom resources hinders my efforts to differentiate lessons and provide meaningful learning experiences for all students. (Ab Hajis & Othman,2025; Santos & Lim,2021)				
4. The absence of collaboration, mentoring, and peer support among teachers reduces my confidence and motivation to apply differentiated instruction in the classroom. (Suryati et al., 2023)				
5. Insufficient or unclear administrative policies, guidelines, and frameworks make it challenging to understand how differentiated instruction should be implemented in my school. (Letzel et al.,2025)				
6. I feel demotivated to use differentiated instruction when there is little recognition, encouragement, or support from school leaders and administrators.				

(lavania & Nor,2020' Indrawati et al., 2024)				
7. Limited support in terms of planning time, reduced workload, or collaborative planning sessions prevents me from effectively preparing and delivering differentiated lessons. (Rohaeti et al., 2020; Indrawati et al.,2024)				
8. Teachers in rural or under-resourced schools face greater challenges in implementing differentiated instruction due to inadequate school infrastructure, classroom facilities, and instructional materials. (Santos & Lim, 2021; An Hajis & Othman,2025)				
9. Lack of continuous mentoring, coaching, or professional guidance affects my ability to refine and improve my differentiated instruction practices over time. (Matiyenga et al., 2024; Bondie et al., 2019)				
10. Limited school support, funding, and resources make it challenging to sustain differentiated instruction practices consistently over time. (Indrawati et al., 2014; Letzel et al., 2025)				

Heavy Workload	**4 Strongly Agree (SA)**	**3 Agree (A)**	**2 Disagree (D)**	**1 Strongly Disagree (SD)**
1. My heavy workload, including multiple teaching responsibilities, limits the time I can dedicate to planning and preparing lessons that are differentiated to meet the diverse learning needs of my students. (Jomuad et al., 2021; Christogonus-Anyanwu& Imiruaye,2024)				

2. Administrative tasks, reporting requirements, and paperwork take valuable time away from developing and implementing differentiated instruction strategies in the classroom. (Marshall et al., 2022; Asman et al.,2023)				
3. High teaching loads, combined with extracurricular duties, reduce my ability to provide individualized attention and support to students with different learning abilities. (Gavin et al.,2021; Shareefa,2020)				
4. Excessive workload contributes to stress and burnout, making it difficult to consistently sustain differentiated instruction practices over time. (Jomuad et al., 2021; Szabo et al., 2022)				
5. Teacher shortages and increased class responsibilities significantly increase my workload, making the effective application of differentiated instruction more challenging. (Asman et al., 2023, Baraily & Sherpa,2024)				
6. Limited time for lesson preparation prevents me from modifying content, instructional activities, and assessment methods to address the diverse learning needs of my students. (Kanwal et al., 2023; Mengistie,2020)				
7. The additional responsibility of managing classroom technology (e.g., projectors, learning management systems, or student data systems) adds to my workload and reduces the time available for instructional planning. (Inegbedion et al., 2020)				
8. A lighter workload and better time management support would allow me to focus more on applying differentiated teaching strategies and engaging all				

students effectively. (Rose & Sika,2019; Szabo et al.,2022)				
9. Excessive workload limits my ability to participate in professional development, workshops, or training opportunities related to improving differentiated instruction practices. (Hajis & Othman,2024; Ali,2024)				

APPENDIX C

C-1. Map of the Research Locale

Figure 1: Calixto D. Montinola National Highschool

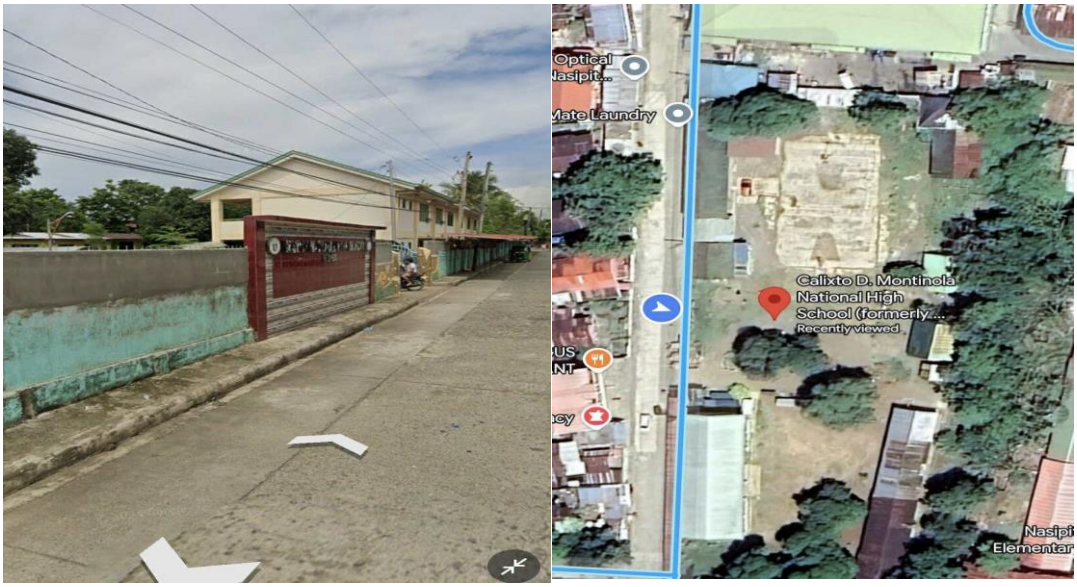


Figure 2: Culit National Highschool

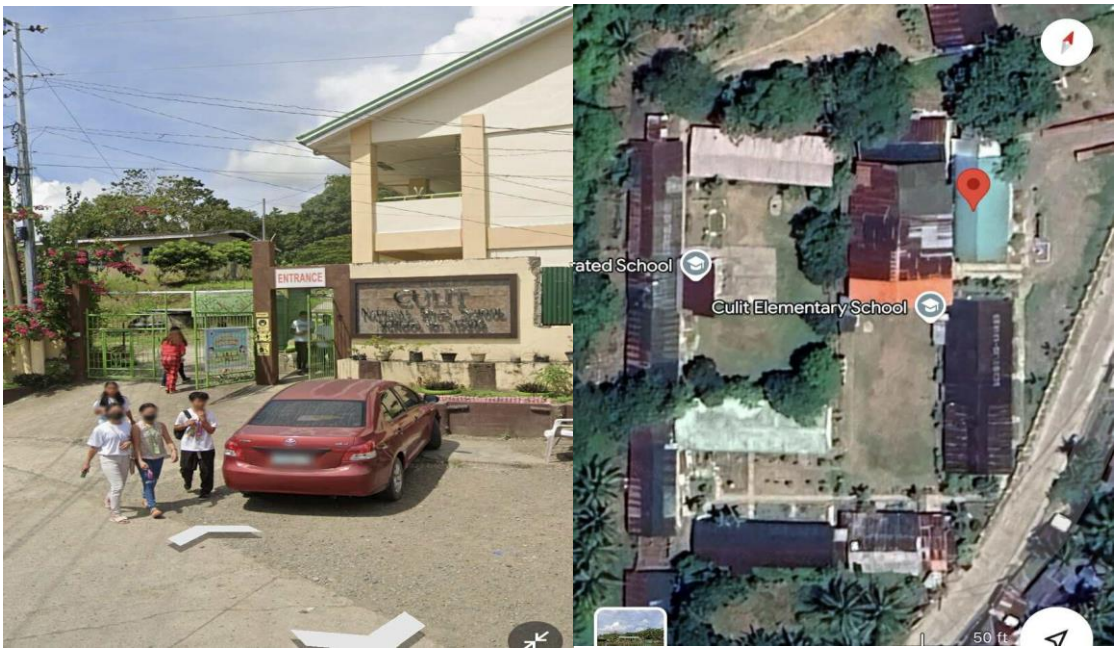
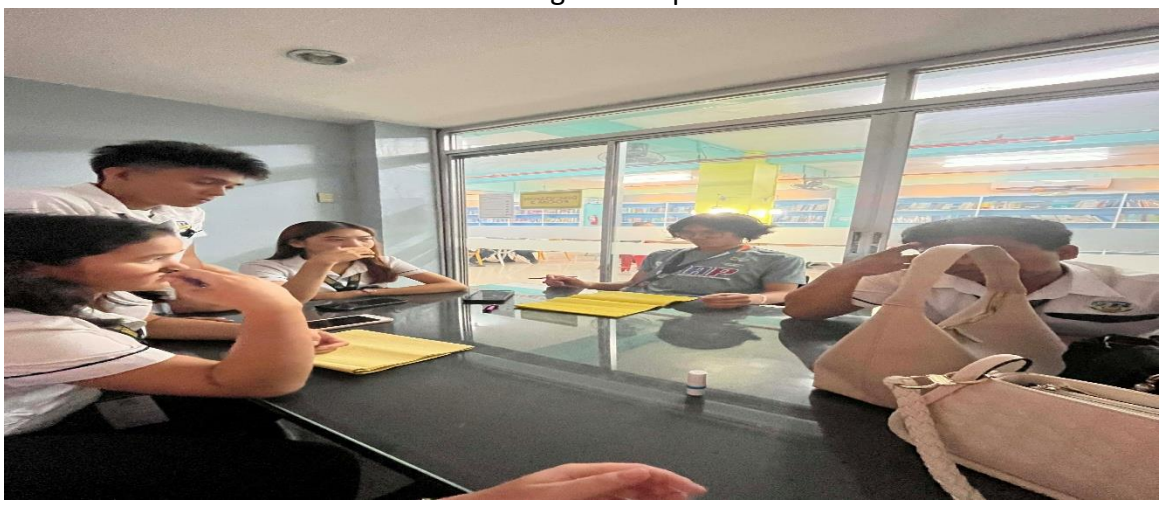


Figure 3: Nasipit National Highschool



APPENDIX D- Narrative & Photo Documentation

Formulating the Chapter 1







Formulating the Chapter 2







APPENDIX E- Certificate of Technology Based Assessment

SAINT MICHAEL COLLEGE OF CARAGA
Atupan St., Brgy. 4, Nasipit, Agusan del Norte 8602, Philippines
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TEST RESULT
TECHNOLOGY-BASED QUALITY ASSURANCE

Date of Test	November 5, 2025	**Control number**	**RBPED125002**	
Research Title	**INFRASTRUCTURAL AND DIFFERENTIATED INSTRUCTION IMPLEMENTATION CHALLENGES OF THE SELECTED PUBLIC SCHOOLS**			
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Test subject	Chapters 1-2				
Total number of words	**17,266**	**Sentences**	**803**	**Characters**	**119,035**

	Test Area	Score	Required	Remarks
	(1) Plagiarism	**98**	**90%**	PASSED
	(2) Grammar	**96**	**95%**	PASSED
	(3) Readability Score	**27**	**25**	PASSED
	(4) AI Text Detection	**24**	**40%**	PASSED
	(5) Reliability *(Test Instrument only)*	N/A	N/A	

FINDING(S) & RECOMMENDATION(S)

Evaluated by:	Approved by:

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